



ODONT 541.1

CARIOLOGY

Dentinal Caries

Lecture #3

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(Tuesday and Wednesday afternoons)

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Quiz Today

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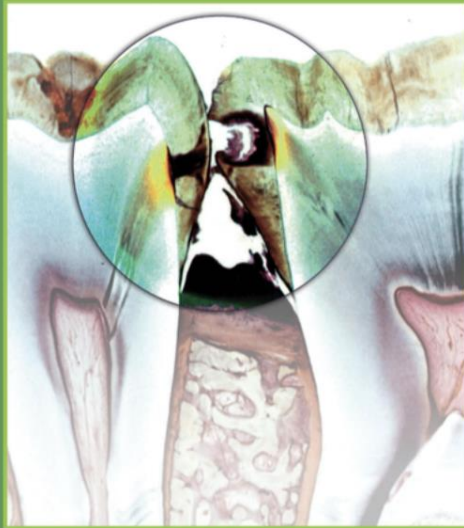
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Recommended Reading

Fourth Edition

Dental Caries

The Disease and Its Clinical Management



Edited by **Ole Fejerskov** and **Bente Nyvad**

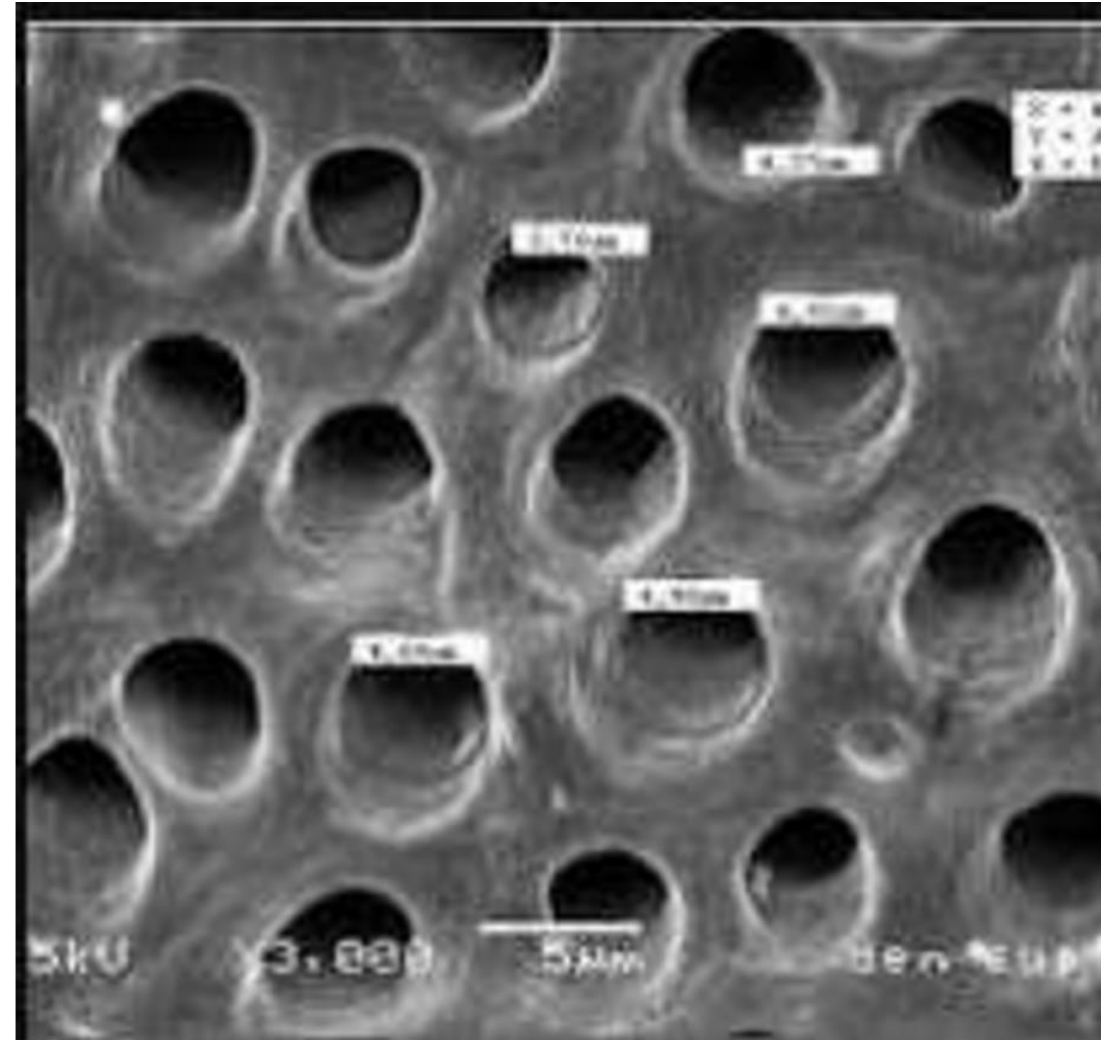
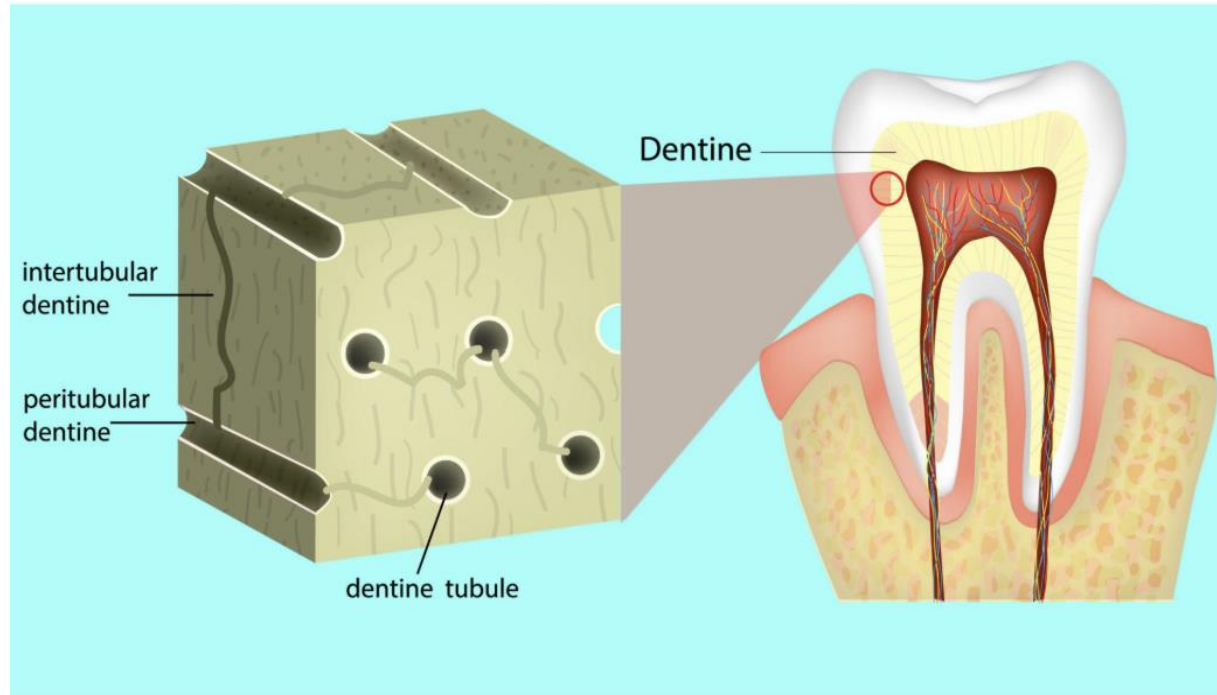
WILEY Blackwell

Textbook:
- Page 221-227

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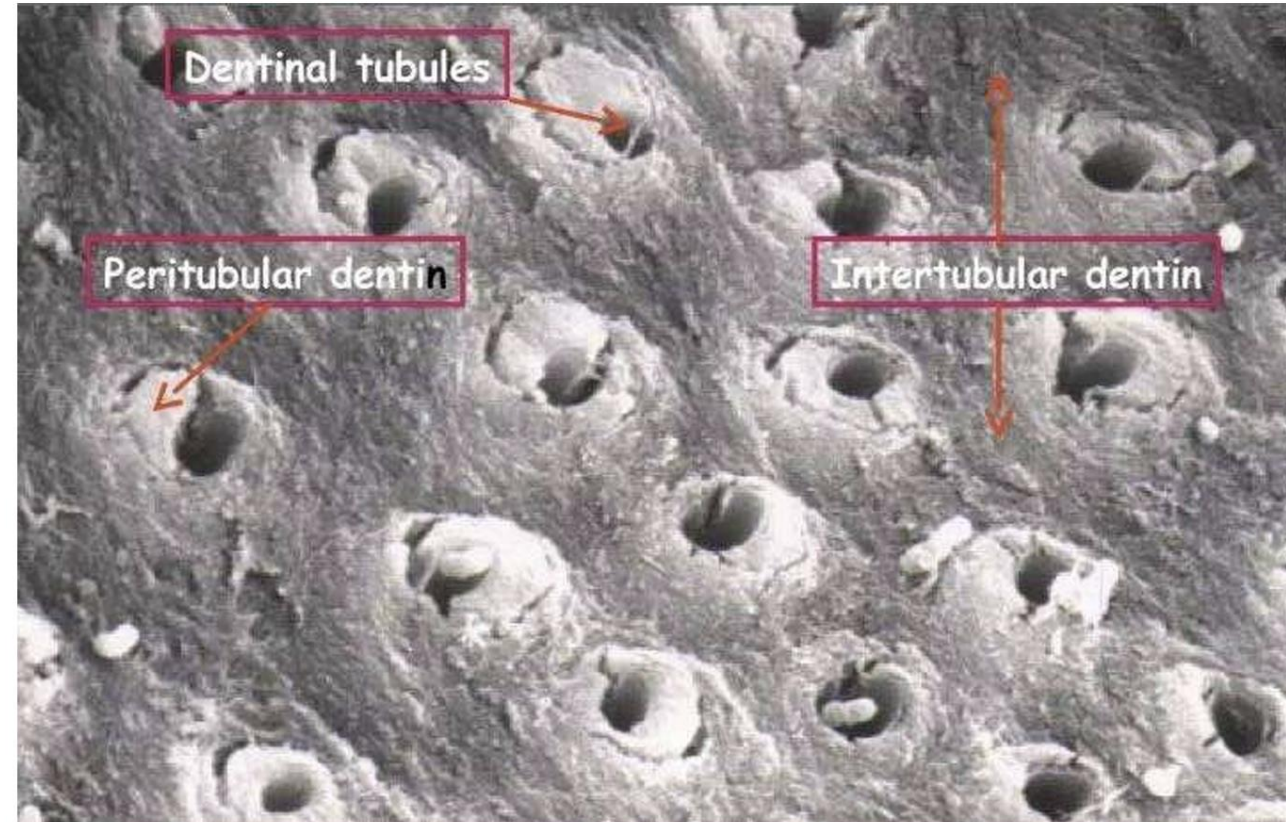
Dentin Structure



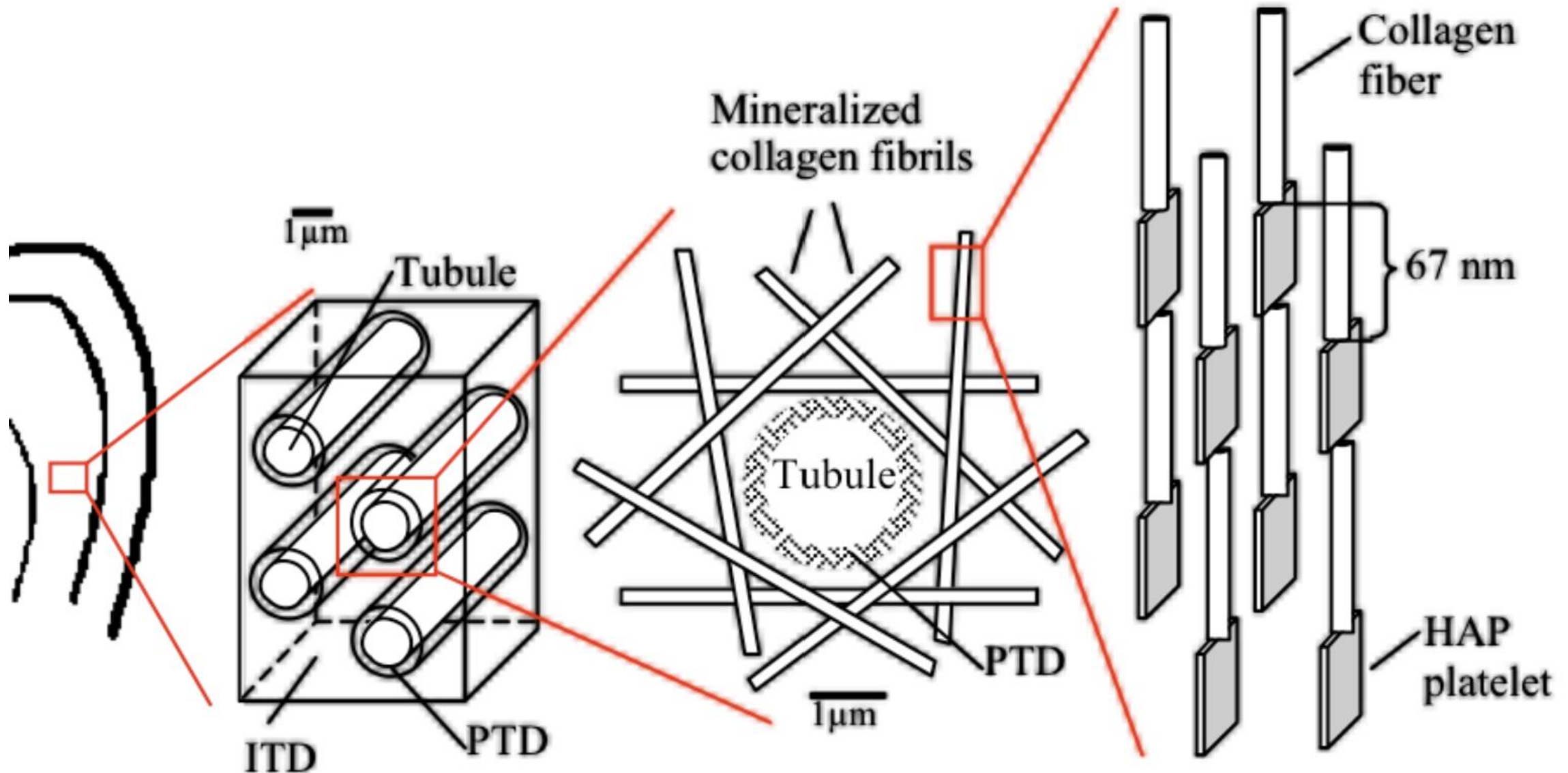
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- Dentinal tubules
- Peritubular dentin
 - o Mineral content 90%+
 - o Encircles each tubule
- Intertubular dentin
 - o Mineral content 70%
 - o Collagen matrix acts as a scaffold between the tubules
 - o Random and irregular arrangement HA among the collagen
 - Smaller size and lack of organized crystalline structure makes intertubular HA more susceptible to demin than enamel HA

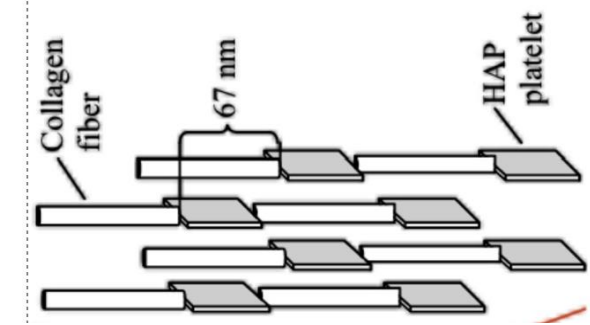


Enamel = large, long, well organized, tightly packed crystals of HA in rods
Intertubular dentin = platelets/fragments of HA interspersed with collagen fibers



Collagen

- 20-30% of dentin is organic material, primarily type 1 collagen
 - Hydroxyapatite demineralizes. Collagen denatures.
- Collagen molecules contain mostly positively charged amino groups
- What is denaturing?



Collagen molecule



Denaturation

Gelatin



Hydrolysis

Peptides



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Dentin Composition

- 50-70 Volume% Mineral
- 20-30 Volume% Organic material
 - Primarily type 1 collagen
- 15 Volume % fluid

Dentin composition varies with the age and history of the tooth

Critical pH of dentin is **6.5**

- Why is dentin more easily demin than enamel?

This means that compared to enamel, caries in dentin:

- Initiates in a more alkaline environment
 - What was normal physiologic pH?
 - Dentin can demineralize even at physiologic pH
 - Is all demineralization caries?
- Progresses faster than in enamel

Dentinal tubules are S shaped in their trajectory from the DEJ to the pulp.

Near the DEJ

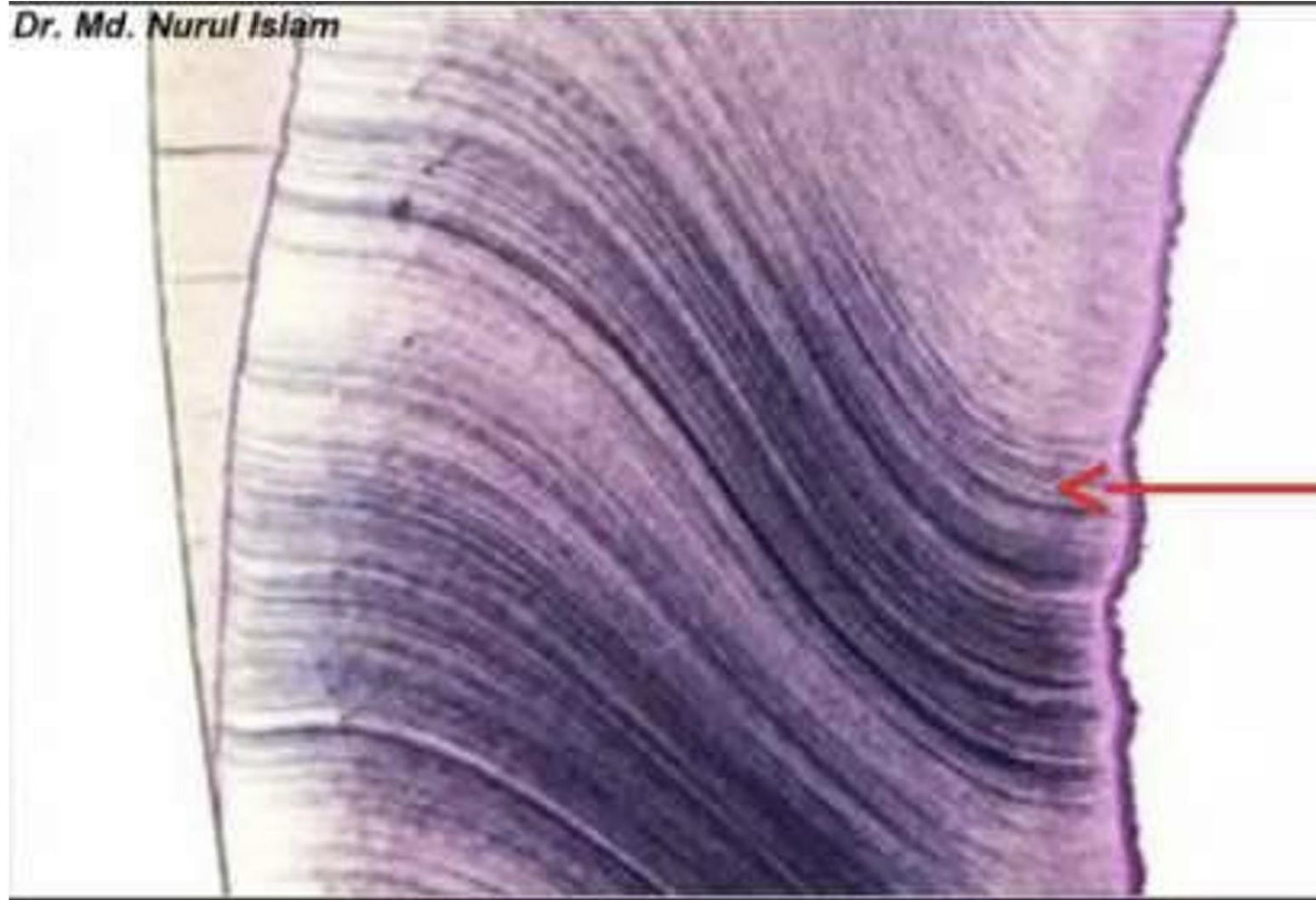
- Noxious stimuli are first received near the DEJ, which is the most superficial layer of dentin
 - Tubules are smaller
 - Tubules are less numerous (25,000 tubules per square mm)

Near the pulp

- Least perception of noxious stimuli
 - Tubules are wider
 - 45,000 tubules per square mm

Therefore, caries progression accelerates toward the pulp

Dr. Md. Nurul Islam



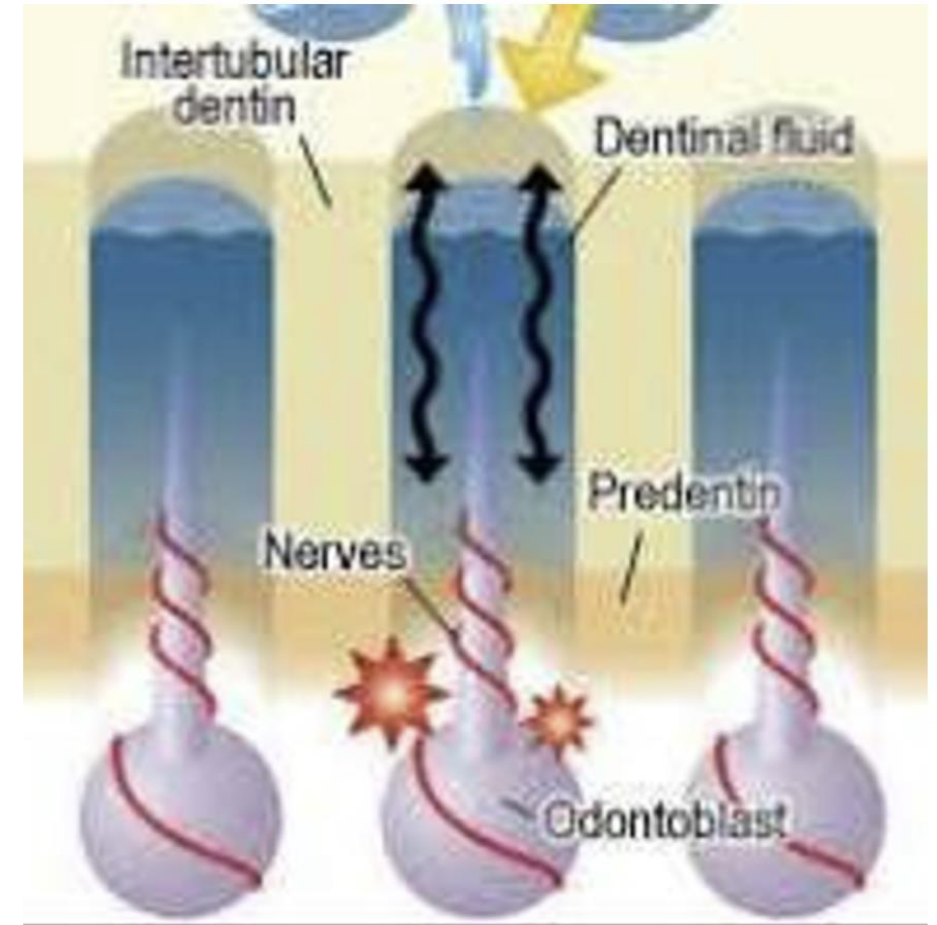
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Within the Dentinal Tubules

- Pulpal fluid
- Odontoblastic process
 - Body of the odontoblast is at the dentin/pulp border.
- Nerve fibers

The presence of odontoblastic processes within the tubules is what makes the dentin a living responsive tissue that feels and can respond to stimuli.

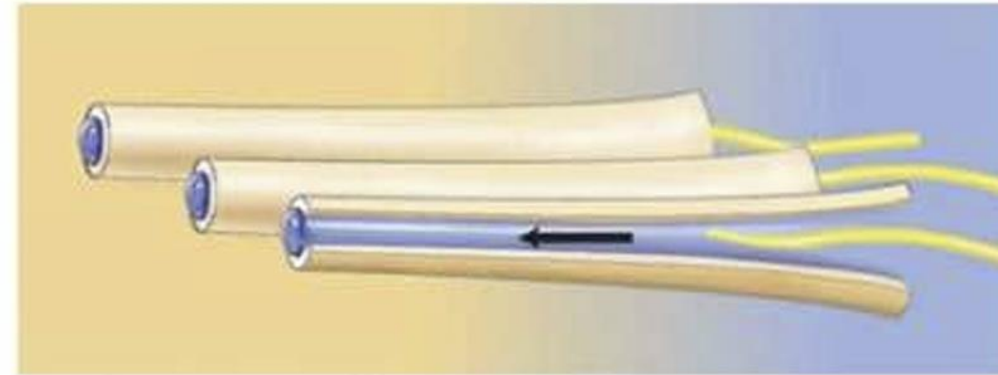


- **Hydrodynamic theory**
 - Movement of pulpal fluid in the dentinal tubules transmits stimulus to the nerves in the pulp

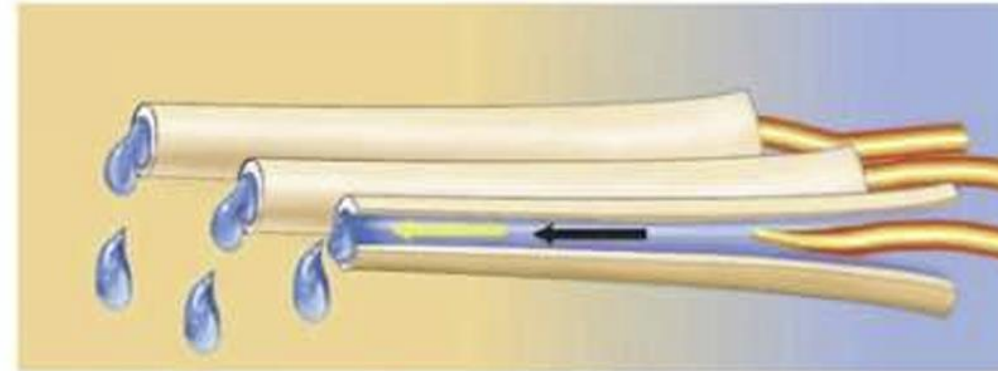
The nerve fibers in the pulp are almost entirely nociceptive.

Therefore, any stimulation of the dental pulp results in pain and only pain.

Why is the dental pulp a pain organ? How is that beneficial? What helpful things can the pulp do when it senses pain?



resting outward pressure



flow in response to stimuli

Figure. The Hydrodynamic Theory illustrated: Fluid within the dentinal tubules can flow in either an inward or outward direction depending on pressure differences in the surrounding tissue. A stimulus that makes contact with a tooth surface with open tubules causes a fluid disturbance, or change in flow direction, which the patient perceives as pain.

The Dentin-Pulp Defense System

1. Tubular sclerosis

- Occlusion of peripheral dentinal tubules with minerals

2. Tertiary dentin formation

- The formation of new dentin as a noxious stimulus gets closer to the pulp

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Desensitizing treatments targeting the hydrodynamic sequence

1. Avoid the pain-producing stimulus

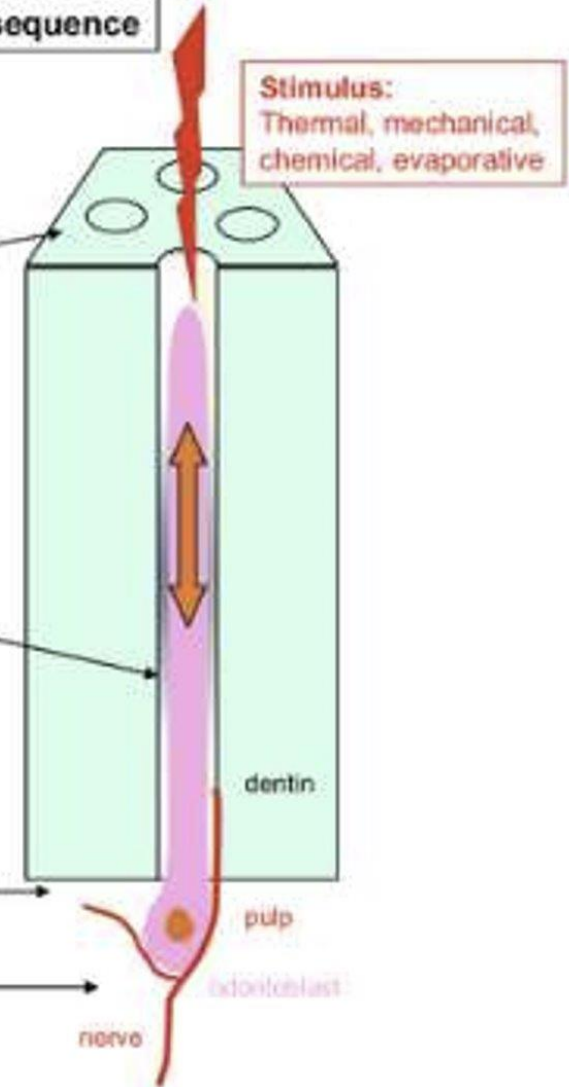
2. Occlusion of the dentinal tubules

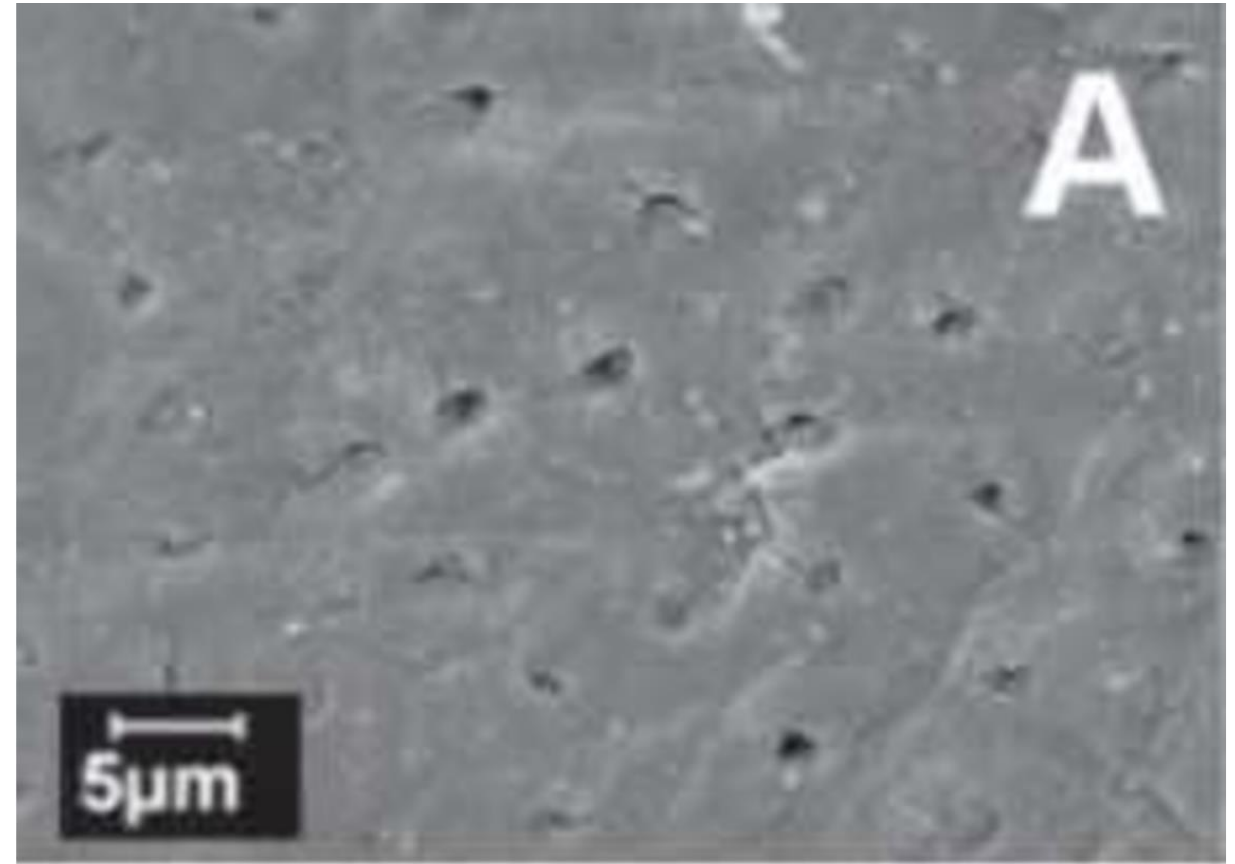
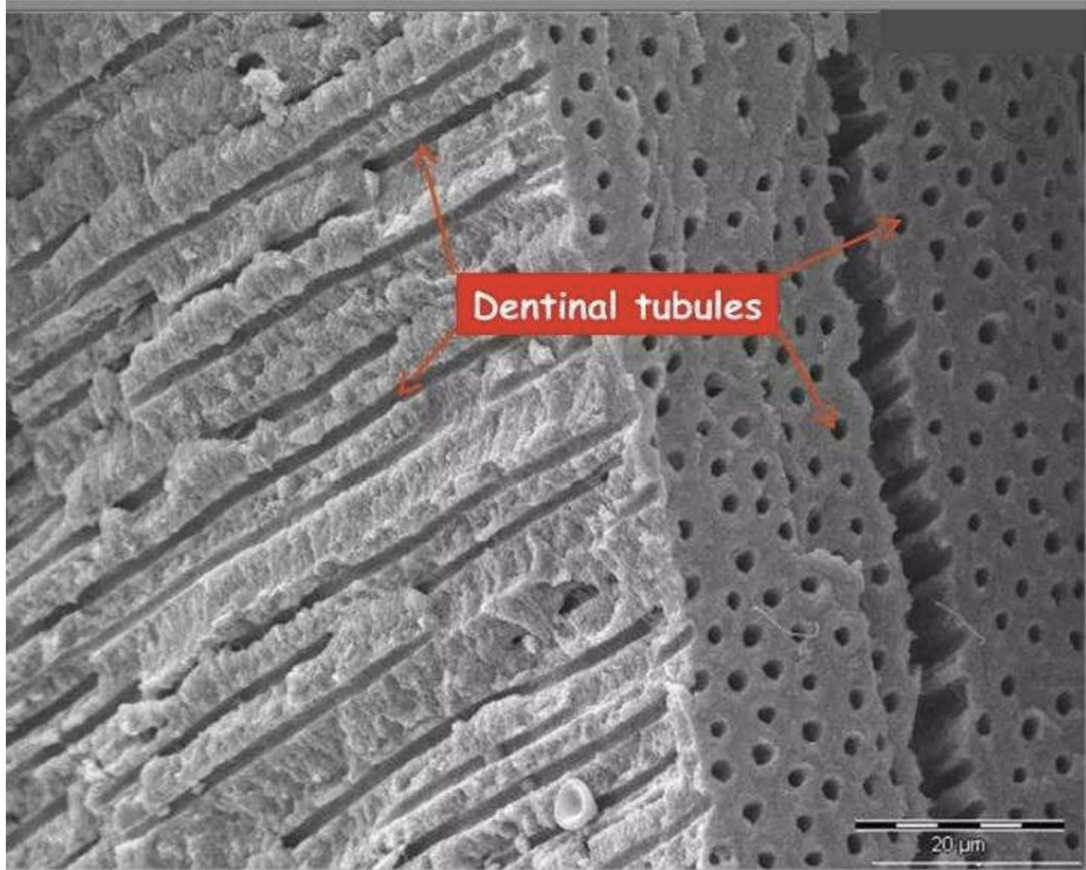
a. Formation of a smear plug (burnishing dentin),
plugging of the tubule opening (application of agents)

b. Increase of formation of the intratubular dentin

c. Induce formation of the tertiary dentin

3. Decrease intradental nerve excitability

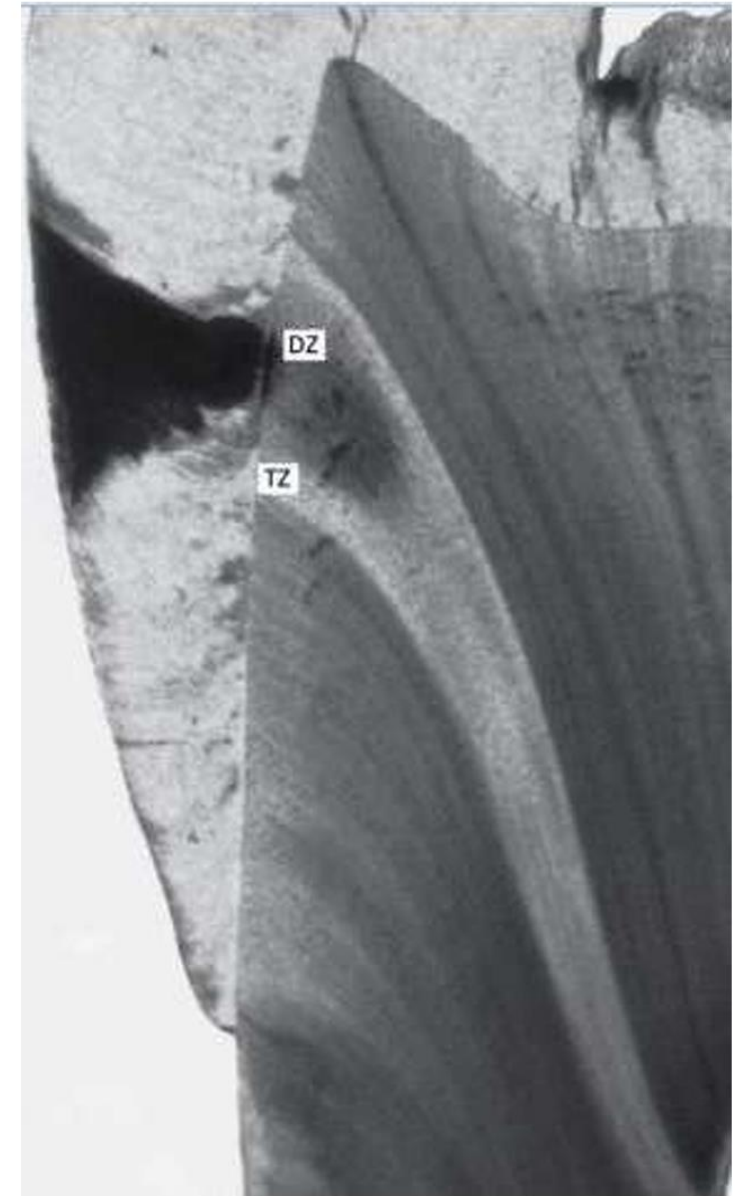




The most common defense reaction by the dentin is tubular sclerosis.

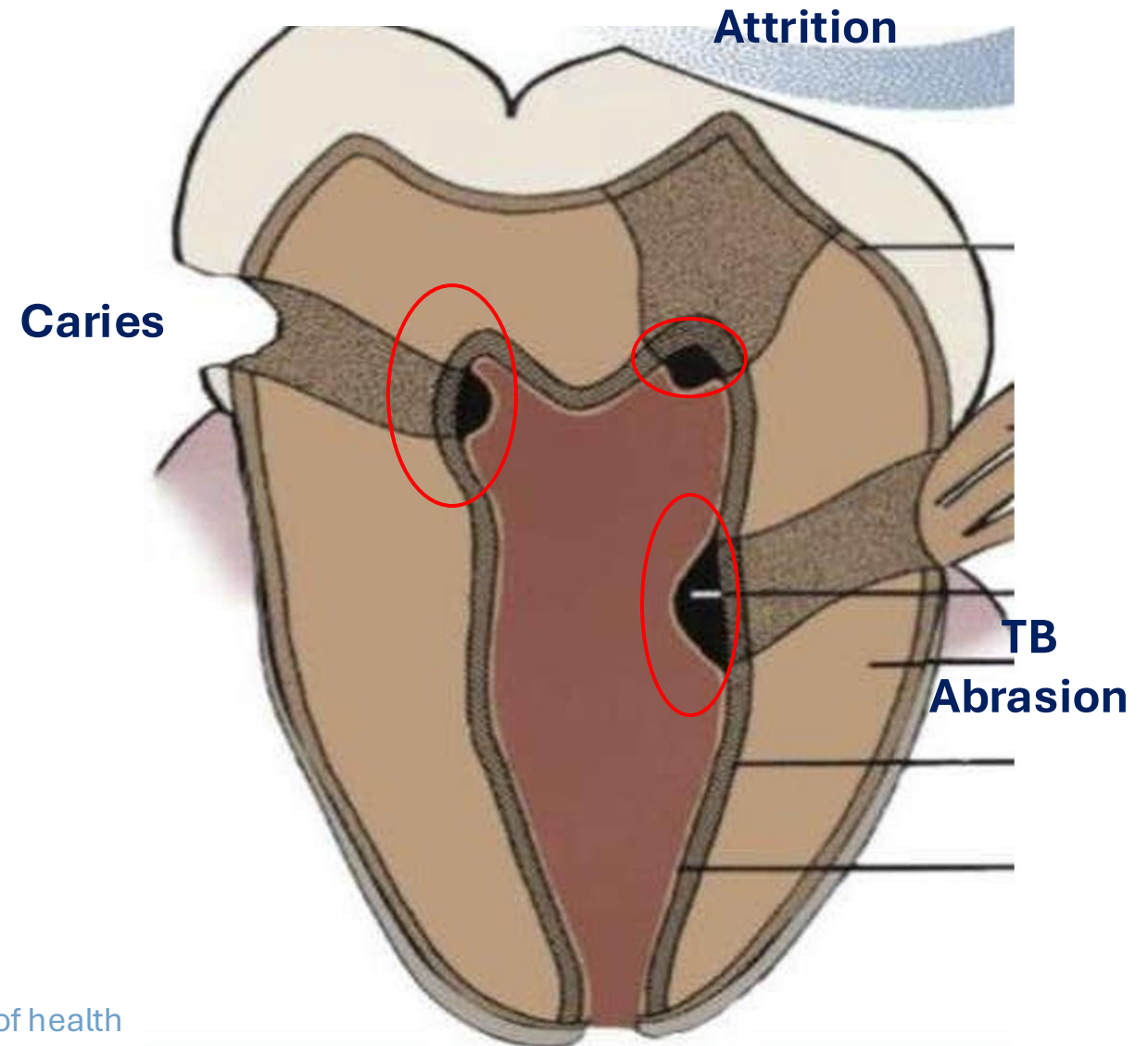
Sclerotic Dentin

- **Sclerotic dentin** (also called **transparent dentin**) is dentin that has undergone **hypermineralization** as a result of a noxious stimulus to occlude the tubules.
 - 90% mineral
 - Makes the sclerotic dentin inside the tubules are more resistant to demineralization.
 - Mineral deposits progressively fill the dentinal tubules, making the peritubular dentin thicker and the tubule narrower.
- Begins at the DEJ and progresses in a pulpal direction.
- Bond strength to sclerotic dentin is typically **25–50% lower** than to normal dentin.
- Notice in this image, the translucent zone is not part of the carious lesion. It surrounds the dentinal lesion.



Tertiary Dentin

- The localized formation of NEW dentin in response to noxious stimuli
- Begins at the dentin-pulp border (deep layers of dentin) and progresses even deeper in the pulpal direction
- Less structured/organized than normal dentin or sclerotic dentin



Types of Tertiary Dentin

Reactionary Tertiary Dentin

- Primary odontoblasts are irritated but not irreversibly injured or destroyed
- The resultant dentin is poorly organized and tubules are sclerotic.

Reparative Tertiary Dentin

- Primary odontoblasts cells are damaged or dead.
- **Replacement** odontoblasts/fibroblasts lay down new tertiary dentin.
- This is the LEAST organized type of dentin and does not have tubules at all.

REACTIONARY DENTINOGENESIS

Mild stimulus or
slowly advancing
caries lesion

Upregulation of
surviving odontoblast
activity and new
tubular dentine
formed

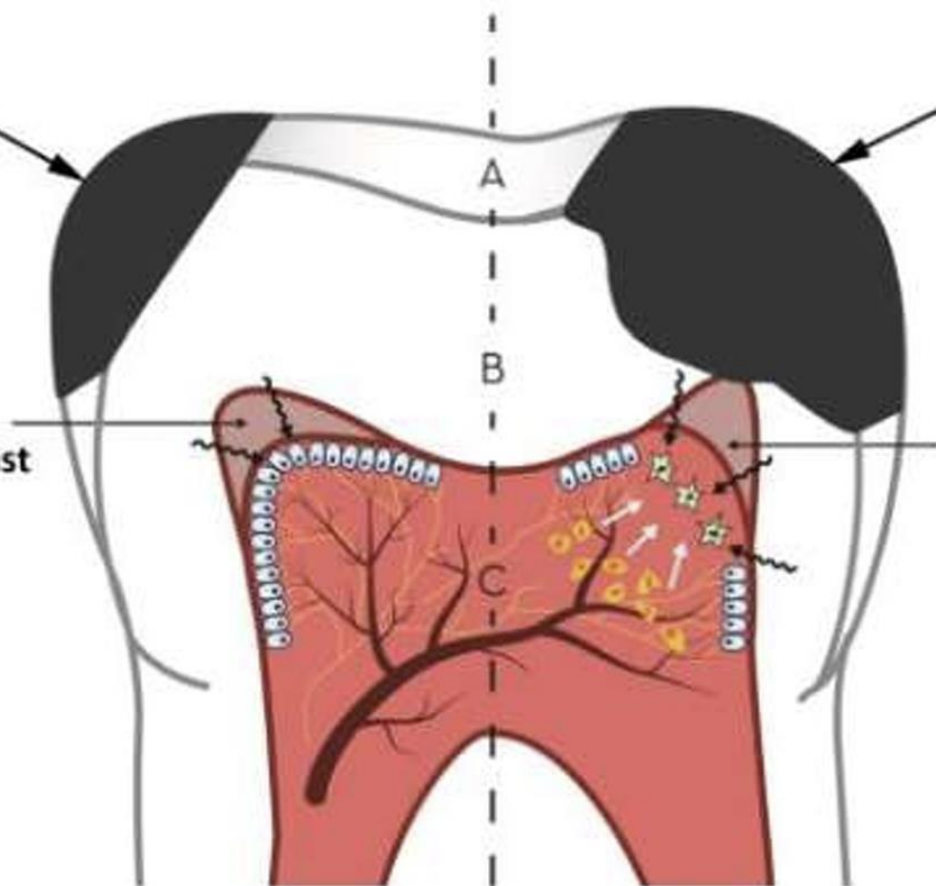
REPARATIVE DENTINOGENESIS

Severe or aggressive
stimulus or rapidly
advancing caries
lesion

Death of primary
odontoblast and
replacement by
odontoblast-like cell

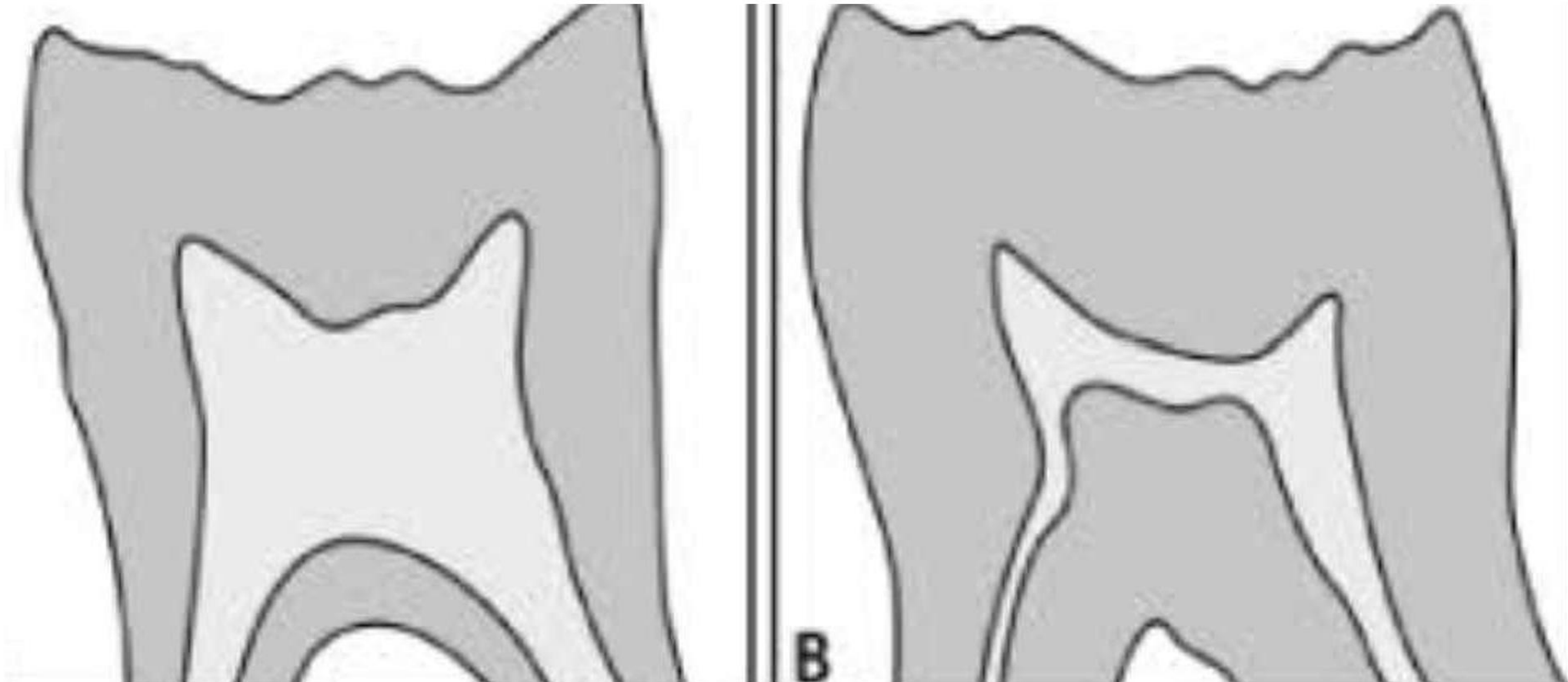
Low quality atubular
mineralised tissue
formed

-  Odontoblast
-  Odontoblast-like Cell
-  Progenitor / Stem Cell
-  Blood Vessel
-  Neuron
-  Dentine Matrix Component



Pulp Recession with Age

Age related pulpal recession is due to **REACTIONARY TERTIARY DENTIN** deposition.



What Does Caries Look Like at the DEJ?

- Initial tubular sclerosis is seen before the advancing front of the enamel lesions even reaches the DEJ
- When contact between the enamel lesion and the DEJ is established, the first signs of dentin demineralization can be seen.
 - The lesion spreads laterally along the DEJ before progressing pulpally.

"Kissing Lesions" pictured

Chalky white = demin enamel

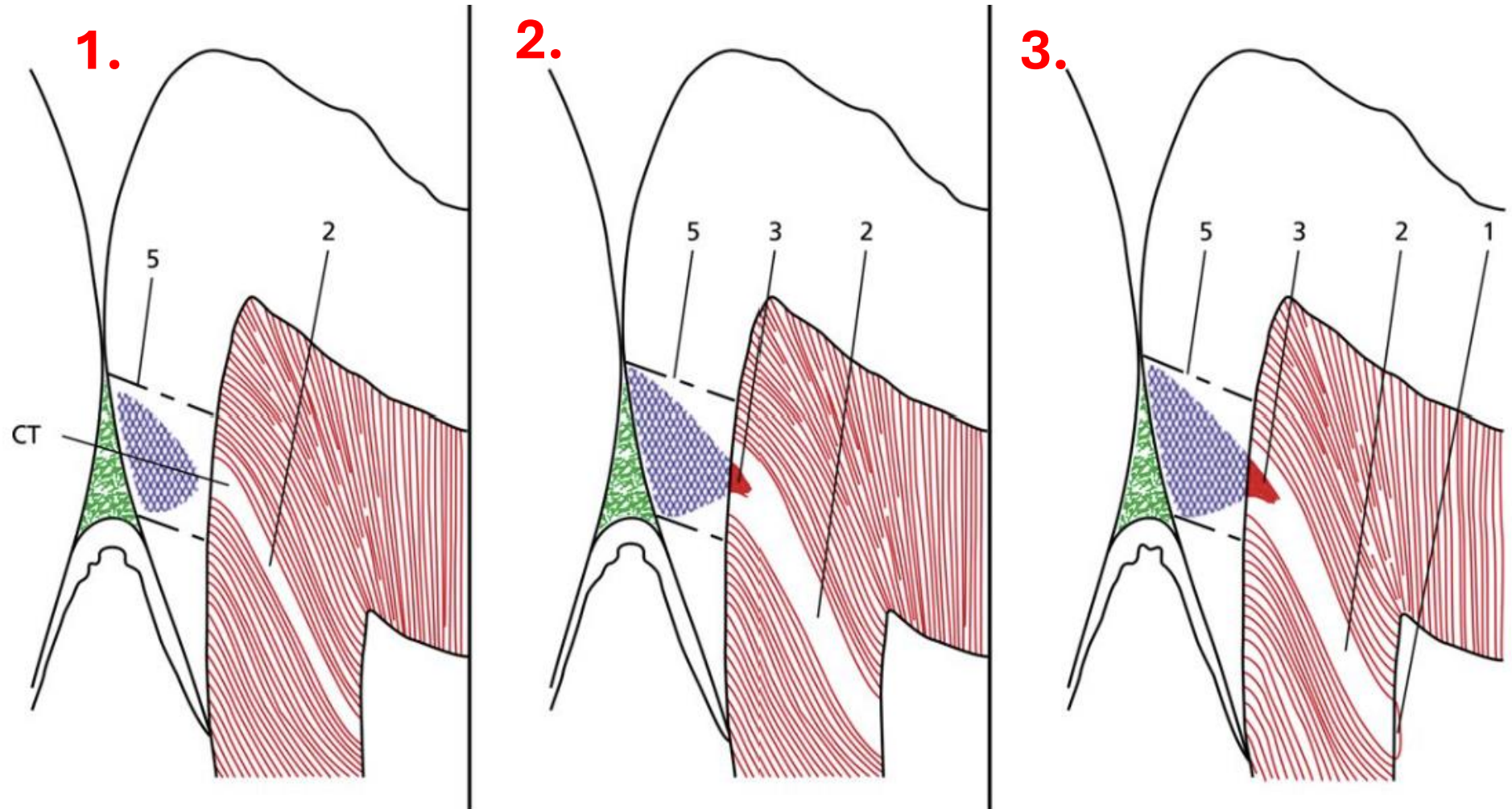
Brown = demin dentin

Light yellow = normal dentin



Cones of Decay: Dentin

Smooth surface:
Pg 65 of textbook



Cones of Decay: Dentin

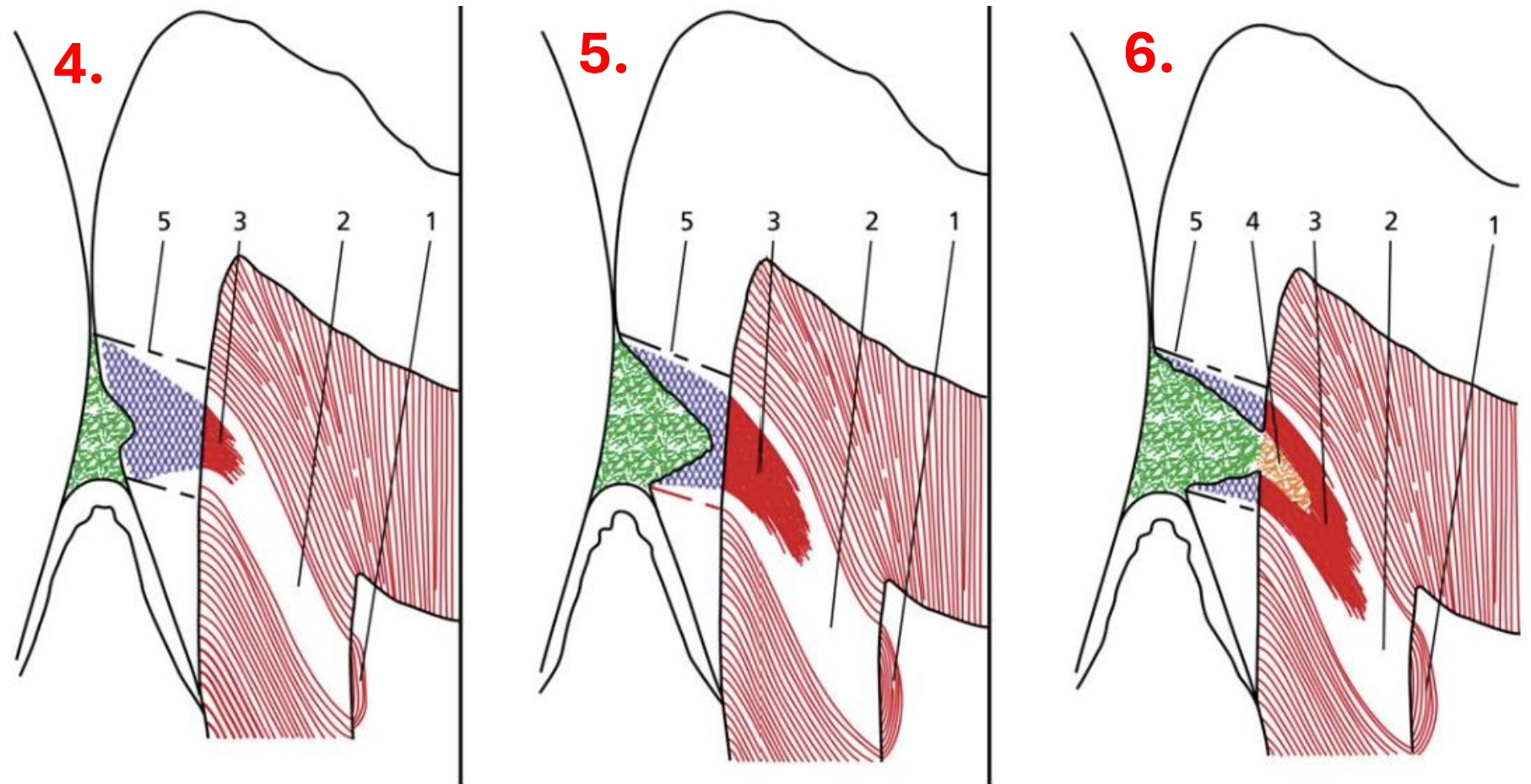


Figure 5.42 Schematic illustration of progressive stages of lesion formation: (1) reactive dentin; (2) sclerotic reaction or translucent (transparent) zone; (3) zone of demineralization; (4) zone of bacterial invasion and destruction; and (5) peripheral rod direction. CT, central traverse line.



Figure 5.57 Ground section of active approximal lesion examined in transmitted light. The triangular enamel lesion reaches the enamel–dentin junction, with demineralization of the outer dentin (ZD) and sclerotic reactions (TZ) corresponding to the less advanced peripheral parts of the enamel lesion.

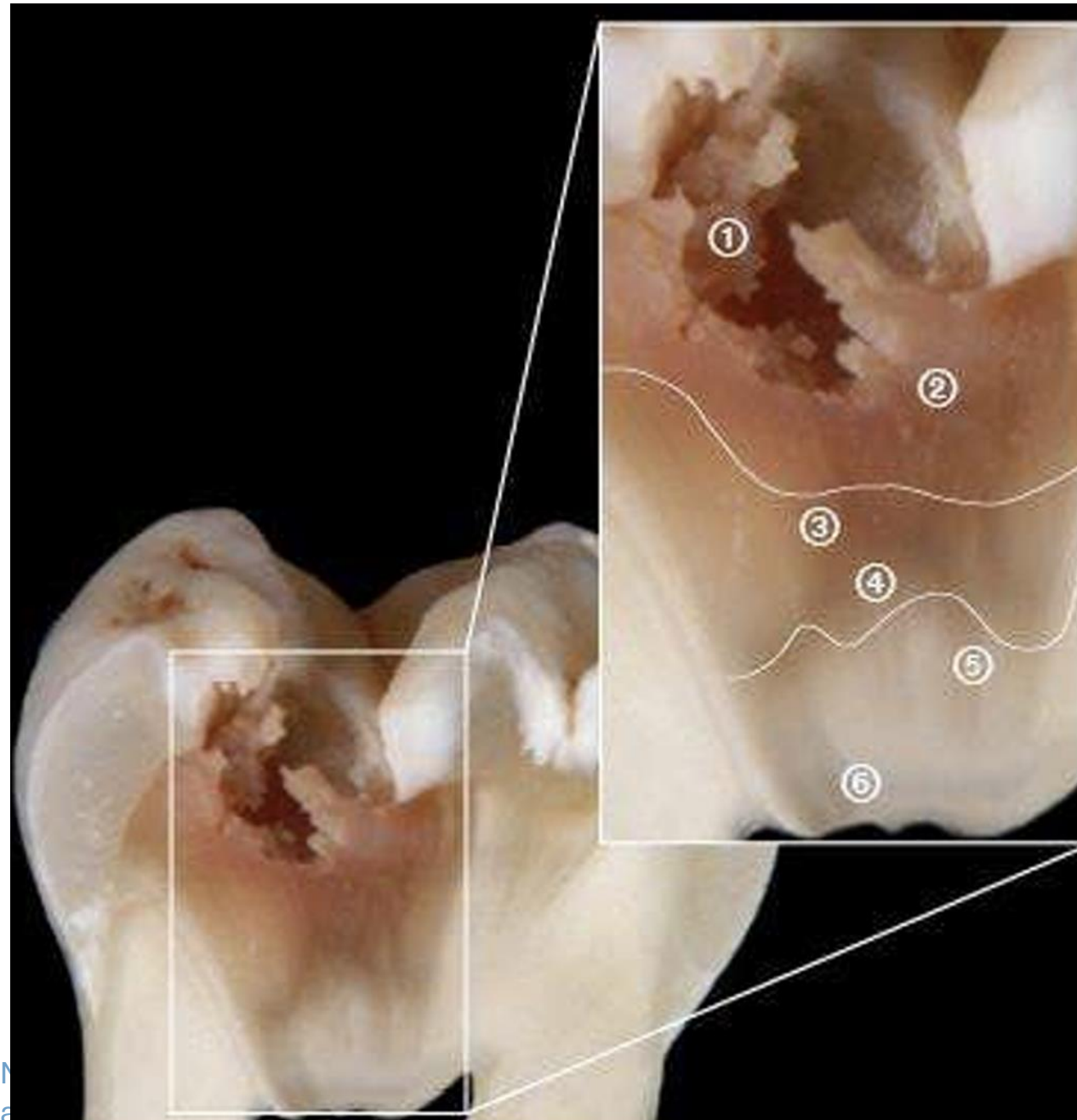


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Histologic layers of
dentin, from
superficial to deep.



Infected Dentin

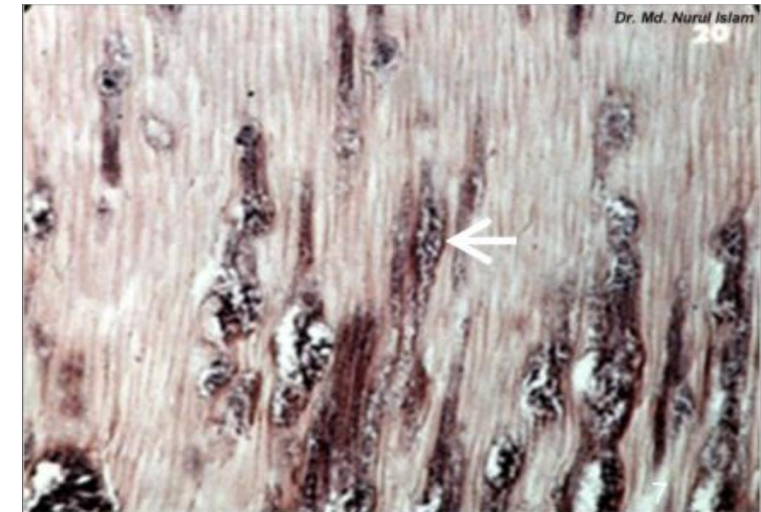
- ① Necrotic zone
- ② Contaminated zone

Affected Dentin

- ③ Demineralized zone
- ④ Translucent zone
- ⑤ Sound dentin
- ⑥ Tertiary dentin

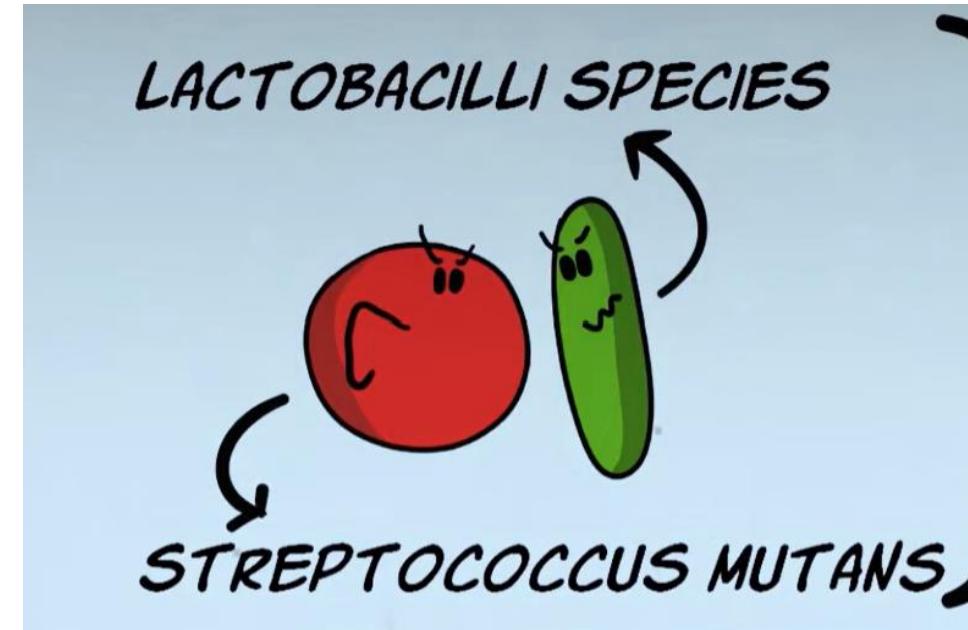
Infected Dentin

- **Contains micro organisms**
- Liquefaction foci:
 - Micro-cavities formed when bacteria digest the dentin matrix from inside dentinal tubules.
- Extensive irreversible degradation/denaturing of collagen matrix due to
 - Bacterial proteolytic enzymes
 - MMPs from the host immune response
- Extensive demineralization of HA due to prolonged exposure to acid
 - Can not be remineralized
 - Loss of tubular structure
- Poor/no bonding
- Soft and necrotic. Must be removed during excavation.



Lactobacillus

- *S. mutans* was important for the initiation of caries. It adhered well to enamel and can process sucrose into glycans (like glucan) that make up an EPS.
- *Lactobacillus* is considered a secondary colonizer and important for the progression of caries.
 - Survives and thrives at pH under 4.5, which is why it isn't highly represented until later in the carious process when the environment has turned more acidic
 - This makes them well suited for deep dentinal lesions, where the environment is very anaerobic and acidic.



Affected Dentin

- **No micro organisms**
- Deeper layer (closer to pulp) than infected dentin
- Collagen is denatured but retains enough structure to support renaturing.
- HA may be demineralized but still retains enough structure to support remineralization.
- Depending on circumstances, could be preserved
 - o Weigh the pros and cons of removing the affected dentin and decide on a case by case basis.
- Compromised by acceptable bonding

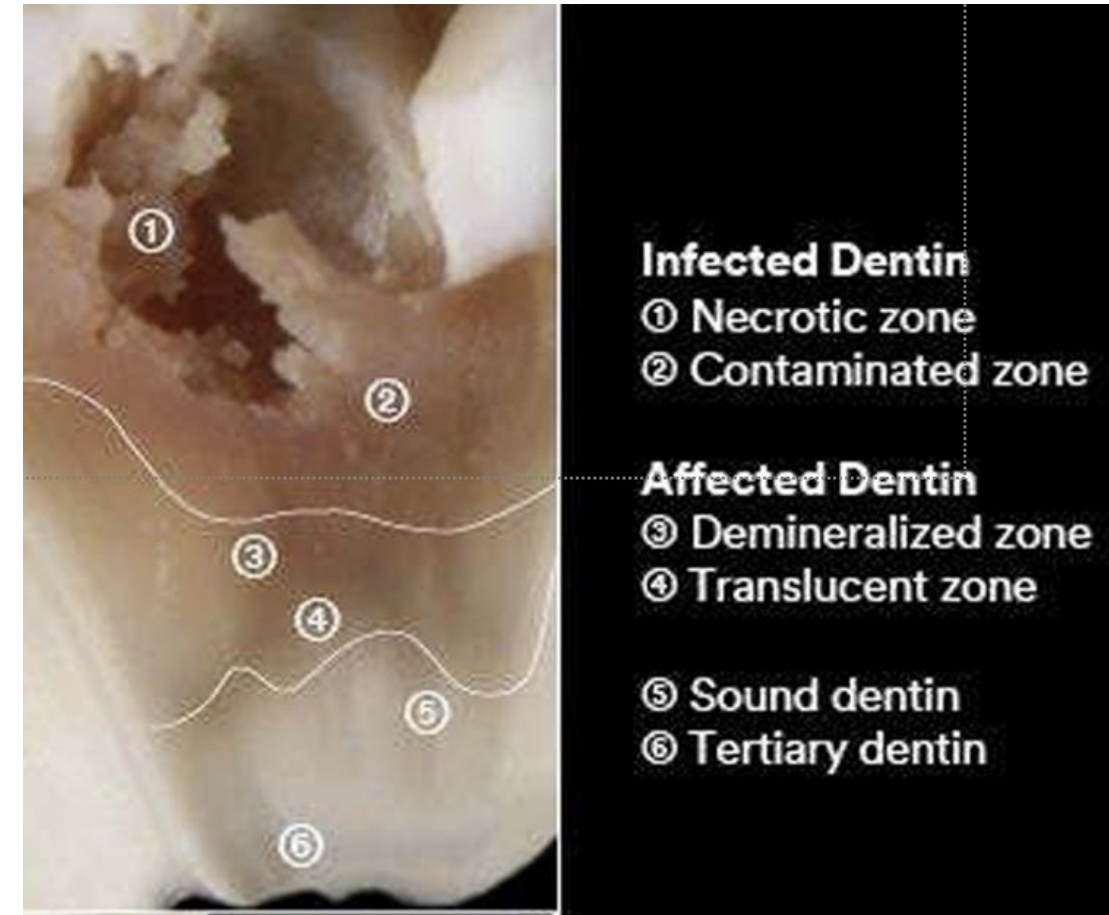




FIGURE 2. *Soft infected dentin with the consistency of cottage cheese.*



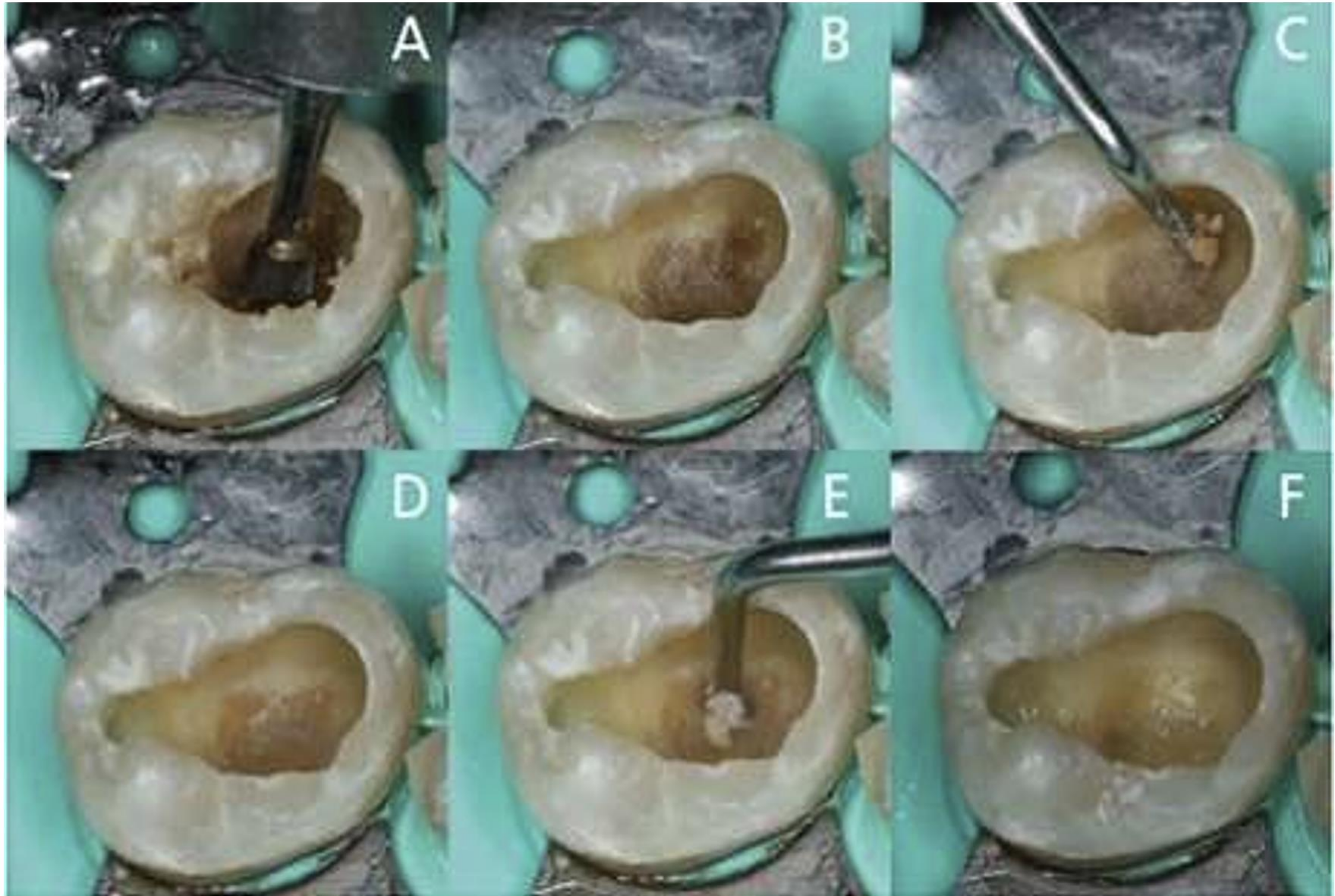
FIGURE 3. *Firm affected dentin with a leatherlike consistency.*

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Do we ALWAYS do
partial caries
removal and leave
the affected dentin
behind?

What are the risks if
we remove too
much? Too little?



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Caries Detector Dye



- Ingredients:
 - 1% dye: highly visible anionic stain that adsorbs to **denatured collagen** and porous organic matrix
 - Propylene glycol: low viscosity solvent that penetrates and carries the dye deeper into dentin
 - Some brands use POLYpropylene glycol instead, which is less viscous and may not penetrate as deep.
- Goal: To stain infected dentin. What is infected dentin?
 - Bacterial infiltration and irreversibly denatured collagen
- Reality:
 - Stains all denatured collagen, reversible and irreversible.
 - Darker color = more denatured collagen.
 - Excavate the dark stain. Leave the light stain.
 - How does this compare to excavate the very soft and leave the leathery dentin?
 - Pink (original) and green behave the same way. But pink stain can be difficult to differentiate if the cavity preparation is contaminated with heme from the gingiva or pulp.
 - Do NOT excavate all the stained tissue
 - So how much should you excavate?
 - Can leave pink stain behind
 - May decrease bond strength

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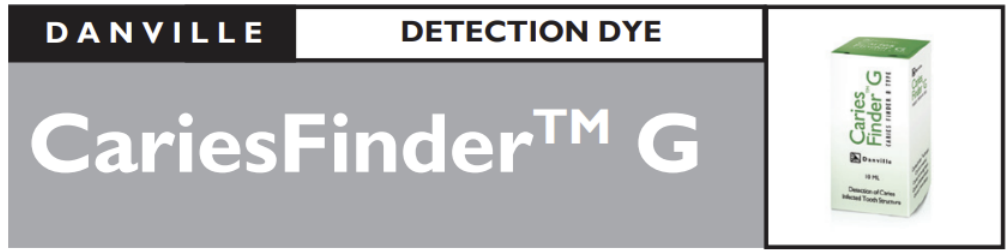
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- Pictured to the right is part of an IFU for a common dye.
 - What are the application instructions?
 - Is the description of what the product does accurate?
 - What is infected dentin?
 - What is affected dentin?
 - What does this solution preferentially stain?
 - Is that infected dentin, affected dentin, both or neither?
- Have you read the IFU for any of the other dental products you have used to date?
 - Have you verified any of those IFUs?
 - Generally speaking, which parts of an IFU are the most trustworthy and which parts are the most in need of verification?
- Does that make this a bad product?
 - It can AID in your personal judgement of how much dentin to excavate, in addition to other factors.

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Caries Finder™ G is a patented green solution in a propylene glycol base. Caries Finder assists the clinician in identifying and removing the outer, unsavable infected dentin while minimizing the loss of inner, savable soft dentin.

Through the extensive research of Dr. Takao Fusayama and other dental researchers, the existence of two distinct layers of soft dentin have been observed. The outer soft dentin is infected, unremineralizable with irreversibly denatured collagen, non-vital and non-sensitive. The inner soft dentin is uninfected, remineralizable with reversibly denatured collagen. Optimally, the outer soft dentin should be removed while the maximum inner non-infected is preserved.

If softness and discoloration are used as the sole guide to caries removal, excessive inner non-infected dentin will necessarily be removed. As Caries Finder only stains the outer infected dentin, Caries Finder provides the clinician with an invaluable guide to help ensure that all the outer infected dentin is removed and the maximum amount of inner dentin is preserved.

APPLICATION INSTRUCTIONS

Caries Finder is generally used in the following manner:

1. Make conservative cavity access.
2. Wash and dry carious dentin.
3. Dispense one or two drops of Caries Finder into a dampen dish and apply to cavity with a brush or sponge.
4. Wait five seconds and rinse with water.
5. Remove green stained outer infected dentin.

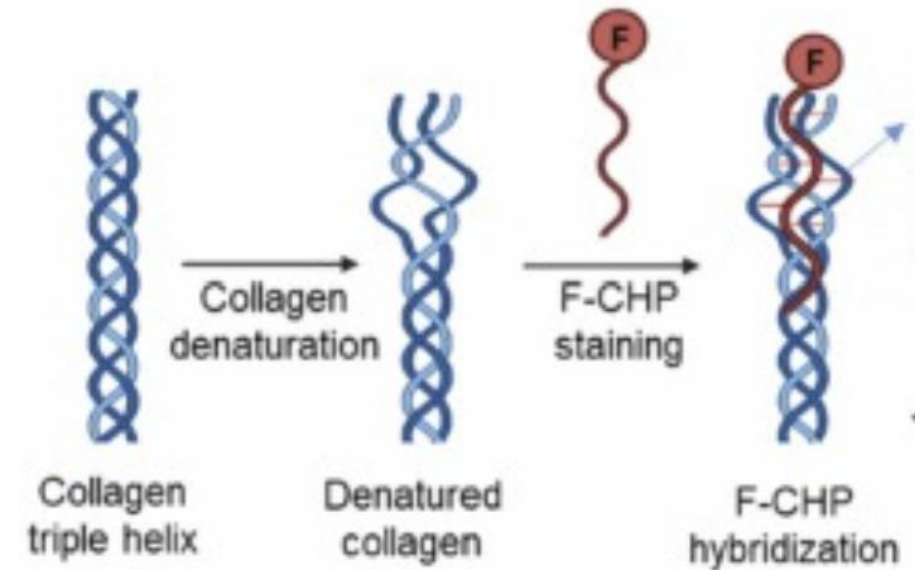
NOTE: When using Caries Finder, the following procedures should be followed to optimize caries removal.

- a. Only remove the non-sensitive green stained tissue. Use care to avoid cutting the sensitive non-stained tissue.
- b. A low-speed rotary instrument should be used for carious dentin removal since the frictional heat generated by the high-speed drill may cause pain before all of the stained outer carious dentin is removed.

Repeat steps 2 - 5 until outer carious dentin is removed and no staining continues.

Caries Detector Dye

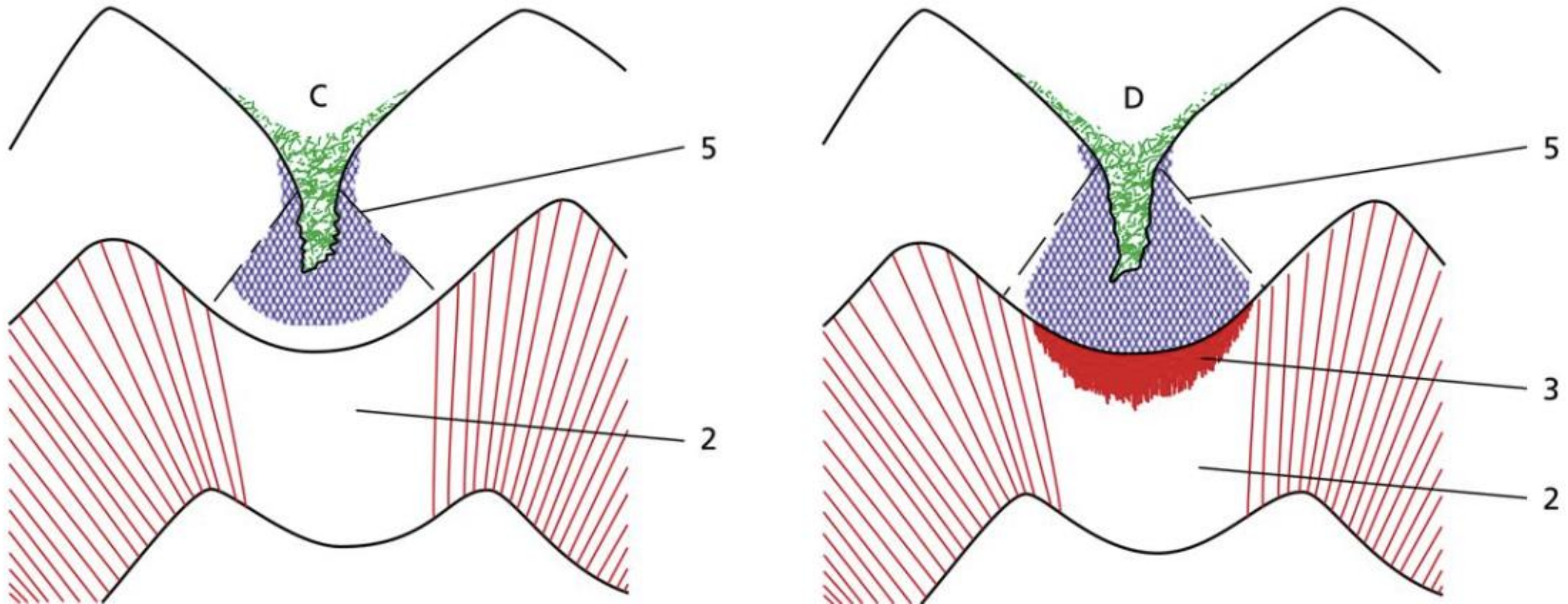
- Sound dentin resists dye because:
 - Collagen is not denatured and is tightly packed (not porous)
 - HA is not demineralized (more dense and less porous)
- Caries detector does NOT tell us:
 - If the stained collagen is reversible or irreversibly denatured
 - If bacteria are present
 - If the carious lesion is active or inactive
 - If you should remove that stained substrate
- Actually, caries detector dyes frequently stain perfectly healthy dentin in two specific anatomic regions.
 - DEJ : naturally porous
 - Circumpulpal dentin: widest and most tubules
- Caries detector dyes were invented in the 1970s. It wasn't until the late 1980s that they realized it only stains denatured collagen (not bacteria, as previously thought).
 - But the IFU I showed you was from this year...



Cones of Decay: Dentin

Pit and fissure: Pg 70 of textbook

How does the progressions of pit and fissure caries differ from the progression of smooth surface caries?



Cones of Decay: Dentin

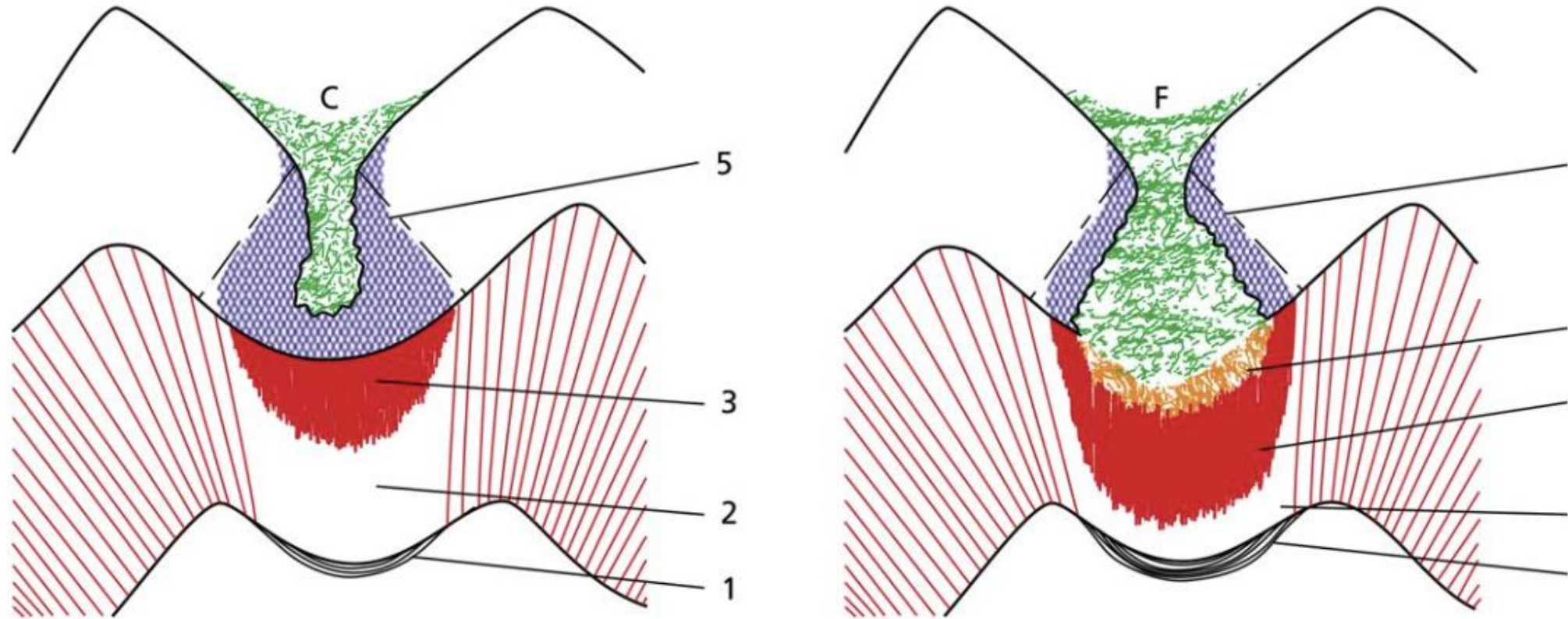
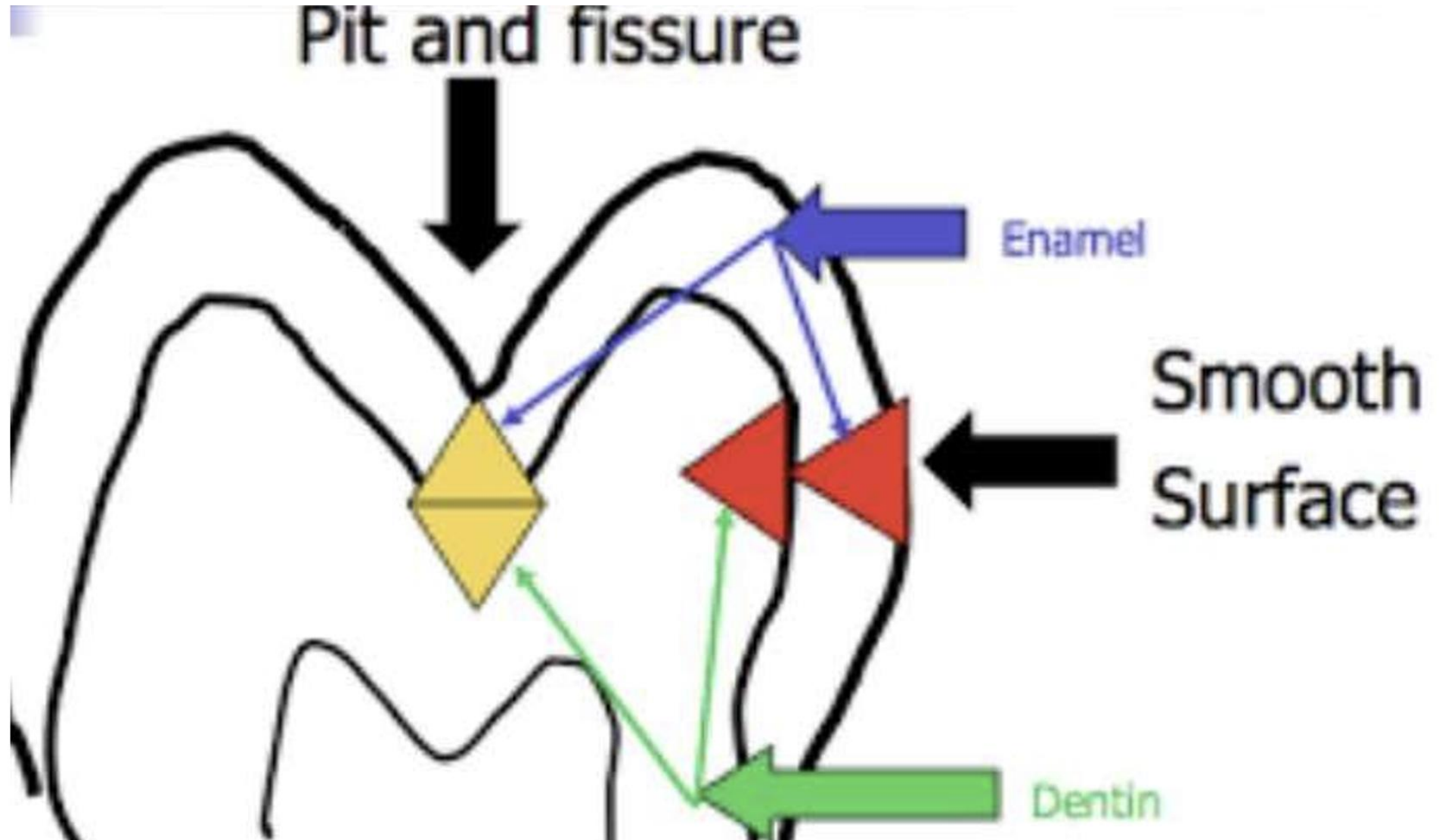


Figure 5.56 Schematic illustration of progressive stages of occlusal lesion formation in an occlusal fossa: (1) reactive dentin in the interface to the pulp; (2) sclerotic reaction or translucent (transparent) zone; (3) zone of demineralization; (4) zone of bacterial invasion and destruction; (5) dotted lines indicate the direction of the prisms.



Notice how much larger the dentinal lesion is than the enamel lesion.

If you clinically examined the occlusal surface of this tooth, you might detect caries. Would you have suspected it's actually this wide and deep??

What would your outline form for a direct restoration of this lesion look like?



Caries Grows Faster in Dentin

Why?

- Less mineral, more organic structures, which are more susceptible to acid attack
- HA in dentin is smaller and less structured
- Tubules allow bacterial infiltration
 - o Bacterial infiltration produces liquefaction foci
- Less oxygen in dentinal environment, favoring growth of anaerobic bacteria
- Less accessible to at home treatments
 - o Less salivary exposure
 - o Harder to brush/floss/pick

What does this mean for the patient?

- Dentists are more likely to recommend intervention in dentinal lesions than enamel lesions
- If no surgical intervention, re-eval dentinal lesions more often than enamel lesions.
- In most cases, non invasive treatments have lower success rates in dentinal lesions than enamel lesions

Examine and interpret every tooth in this image systematically.

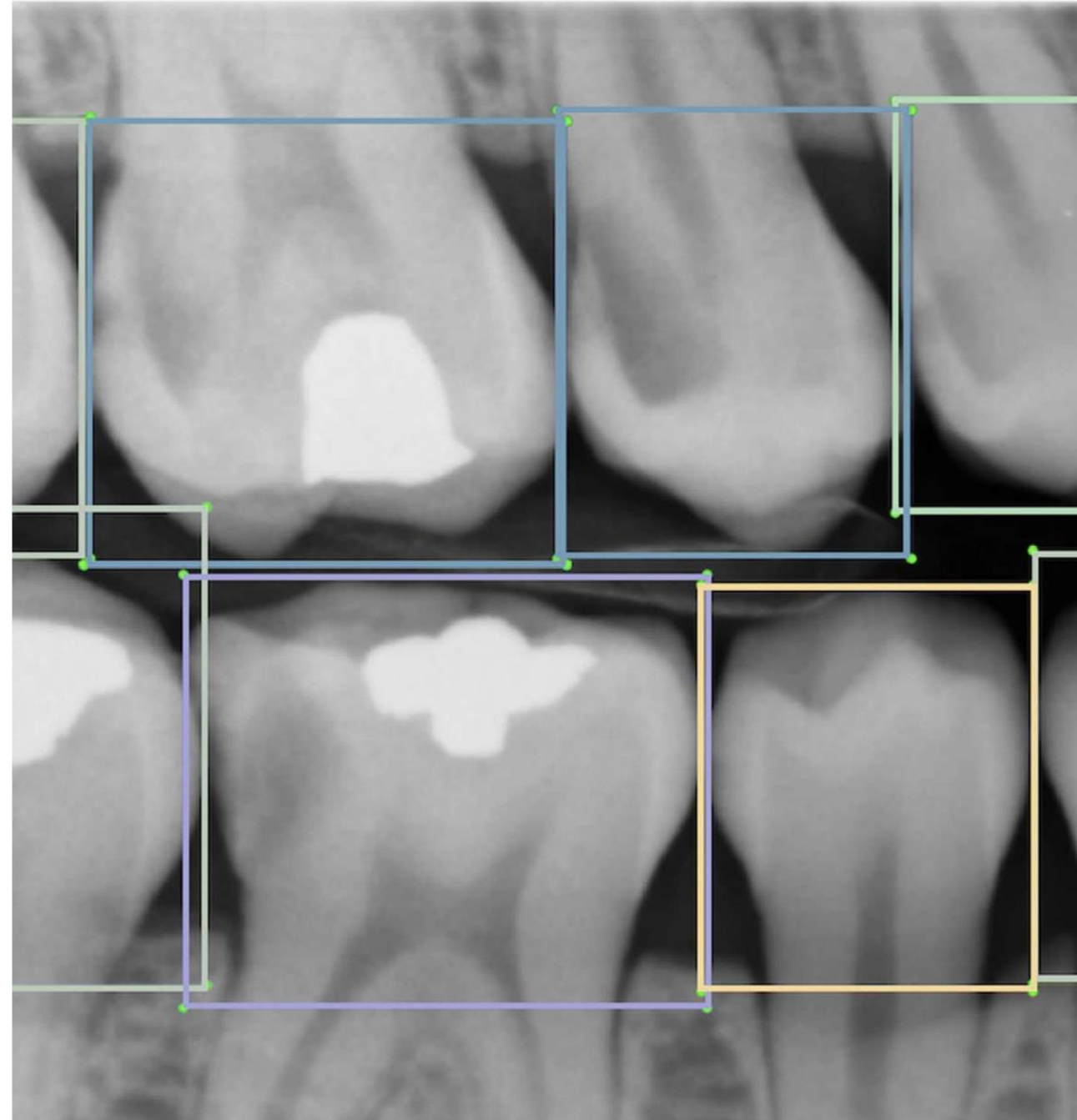
Human error is a leading, pervasive cause of medical malpractice and patient harm, with studies indicating that medical mistakes are the third leading cause of death in the United States.

Approximately 20% to 40% of dental radiographic interpretations contain diagnostic errors.

Systems and checklists are critical for minimizing human error.

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Interpreting Radiographs for Caries (Clinical Crowns Only)

- ☐ What view radiograph is this?
 - Is it of diagnostic quality?
 - Is it recent?
 - Is it relevant?
 - If no to any of the above, what specifically needs to improve on a retake to remediate?
 - Is there radiographic history for comparison?
- ☐ Which teeth are in view?
 - Is that normal? Are any absent? If any are partially visible, is there sufficient suspicion of pathology to attain another image for full visibility?
- ☐ Avoid the urge to zoom in on the obvious tooth! Exercise restraint, systematically examine and interpret each tooth.
 - Suggestion: Go in increasing order #1 -> 32.
 - Suggestion: Go in histological layers, outside in. Enamel -> Dentin -> Pulp
 - Suggestion: Separate radiographic findings into categories.
 - ✓ Existing
 - ✓ Findings
 - ✓ Concerns

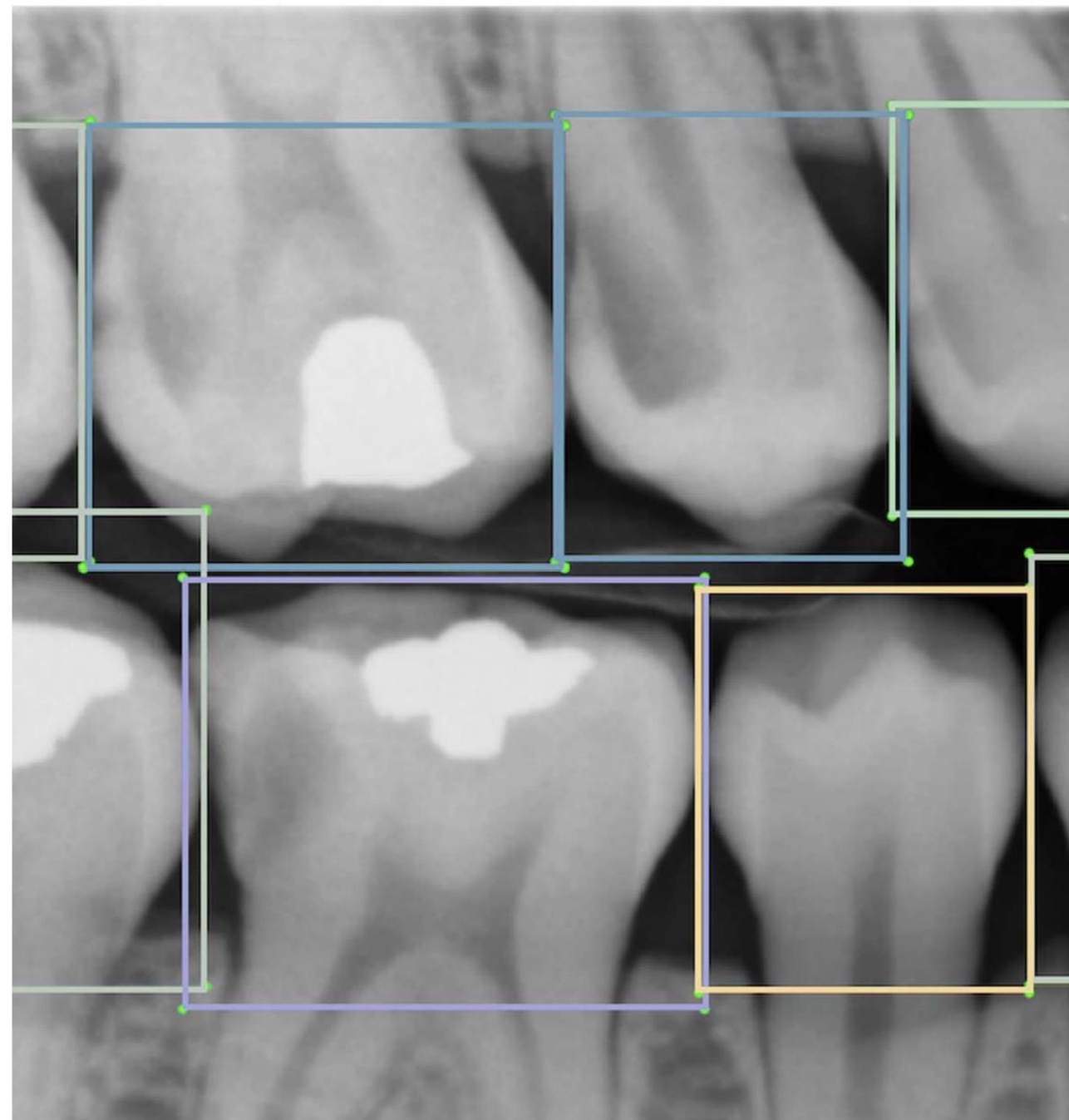
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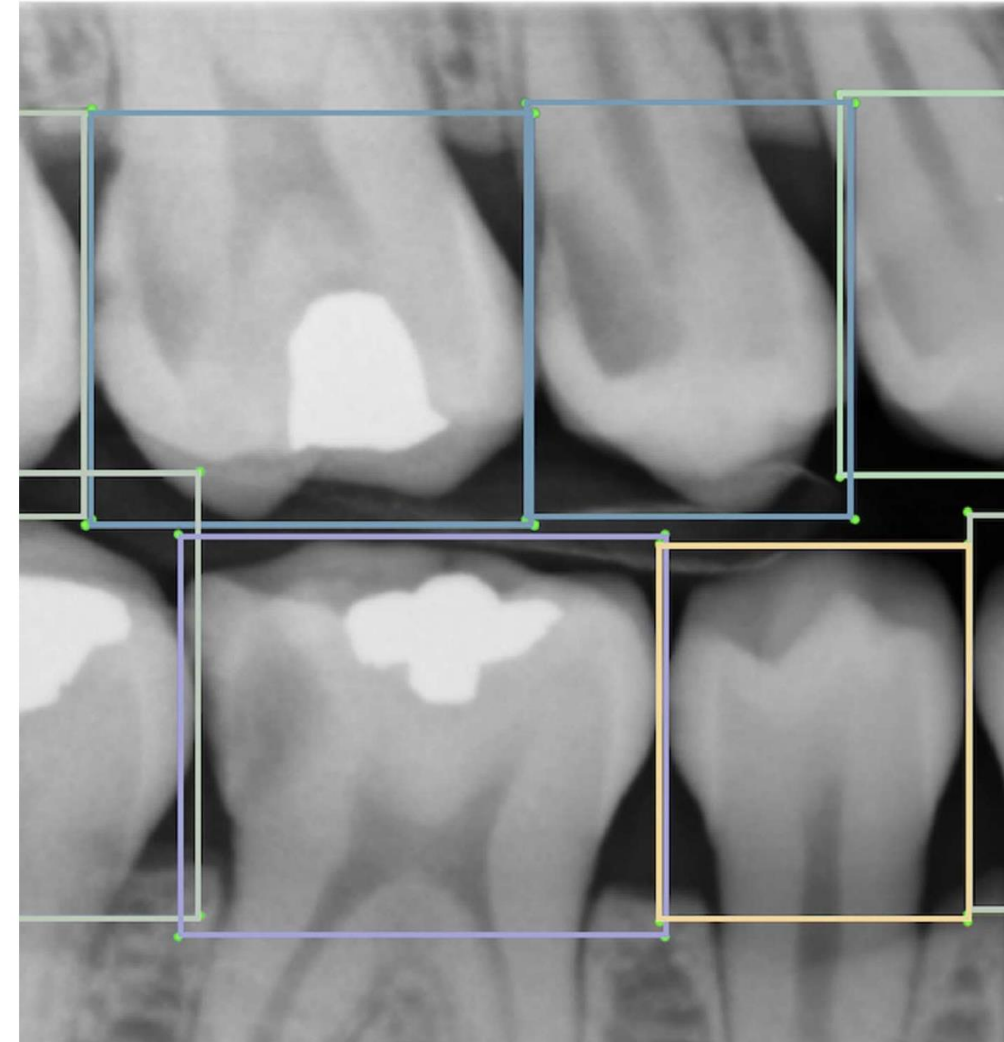
- This is a BW of the right side.
 - It is a premolar view or a molar view?
 - Neither...need to ensure supplementary images complete the diagnostic series
 - Is bone level visible in both arches? Yes
 - Are contacts "open"?
 - At minimum, can see DEJ of all interproximal surfaces.
 - Image integrity?
 - Elongation? Foreshortening? Artifacts? Movement? Other errors?
- Which teeth are in view
 - Partial crowns: #2, 5, 28, 31
 - Complete crowns: #3, 4, 29, 30

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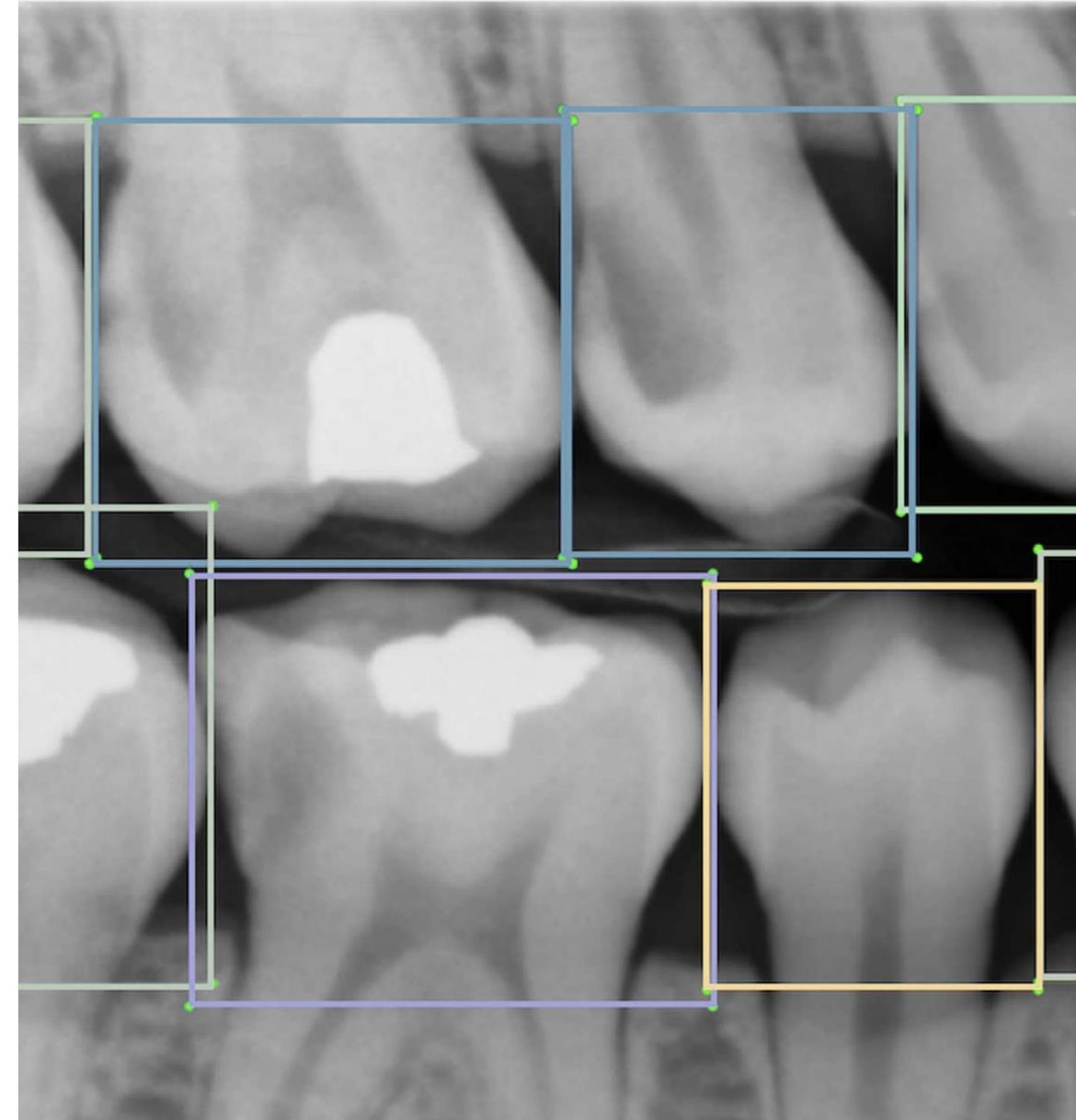
- Enamel
 - Existing restorations
 - What material? Which surfaces or structures involved?
 - How large and deep?
 - Is it radiographically sound? Open margins, overhangs, recurrent caries, broken, etc?
 - Best guess, then verify clinically.
 - Never diagnose on radiographic appearance alone.
 - Radiolucencies
 - Damage?
 - Caries?
 - Would the presence of a carious lesion make sense in this site based on your understanding of the caries process? Is it a protected site? Or did it used to be?
 - Again, which surface and how large and deep?
 - E1/E2
 - Percent of surface area or volume
 - Suspicions
 - Contacting surfaces (kissing lesions)?
 - Cavitation?
 - Clearance
 - Did you look at the WHOLE structure? Not just the suspicious or obvious parts?



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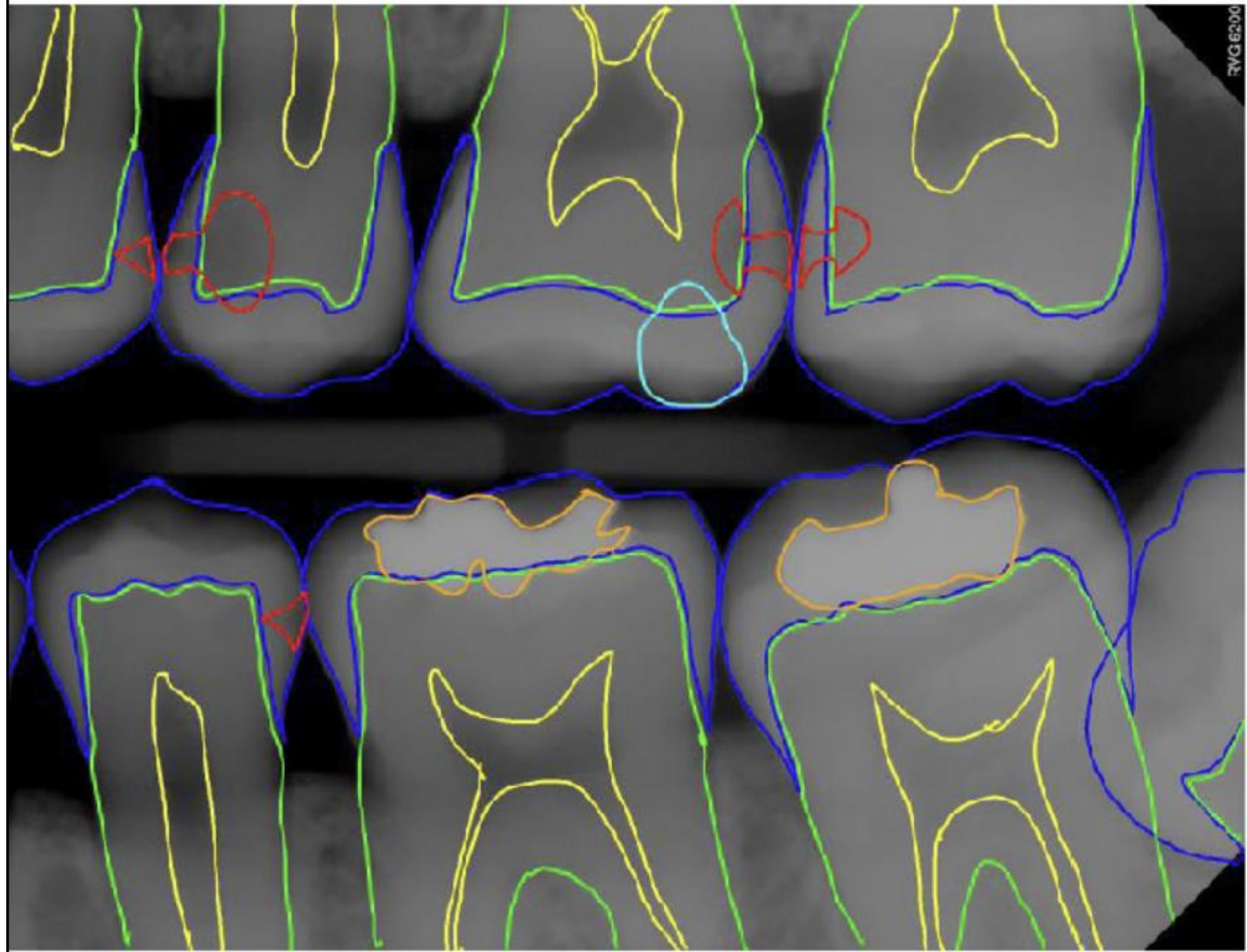
- Dentin and Pulp
 - DEJ
 - **Caries in dentin spreads laterally along the DEJ, establishing a wide base, before it progresses pulpally**
 - **Remember that the rate of caries progression accelerates once the DEJ is reached.**
 - Depths
 - D1/D2/D3
 - Proximity to pulp
 - Pulpal reaction to carious lesion?
 - Is this a good thing or a bad thing?
 - How does it compare to other teeth?
- Do symptomatic teeth look radiographically different than asymptomatic teeth?
- Is the largest lesion typically the most symptomatic one?
- HOW ACCURATELY DO RADIOGRAPHS REPRESENT THE SIZE OF DENTINAL LESIONS?



The use of radiographic screening is rapidly increasing.

However, interpretation is ultimately solely the responsibility of the prescribing dentist.

If you prescribe an image, you must fully interpret the image and document your interpretations (or refer for full interpretation).



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Which type of radiographic image is most diagnostic for caries?

What should you do if you see caries on a PA?



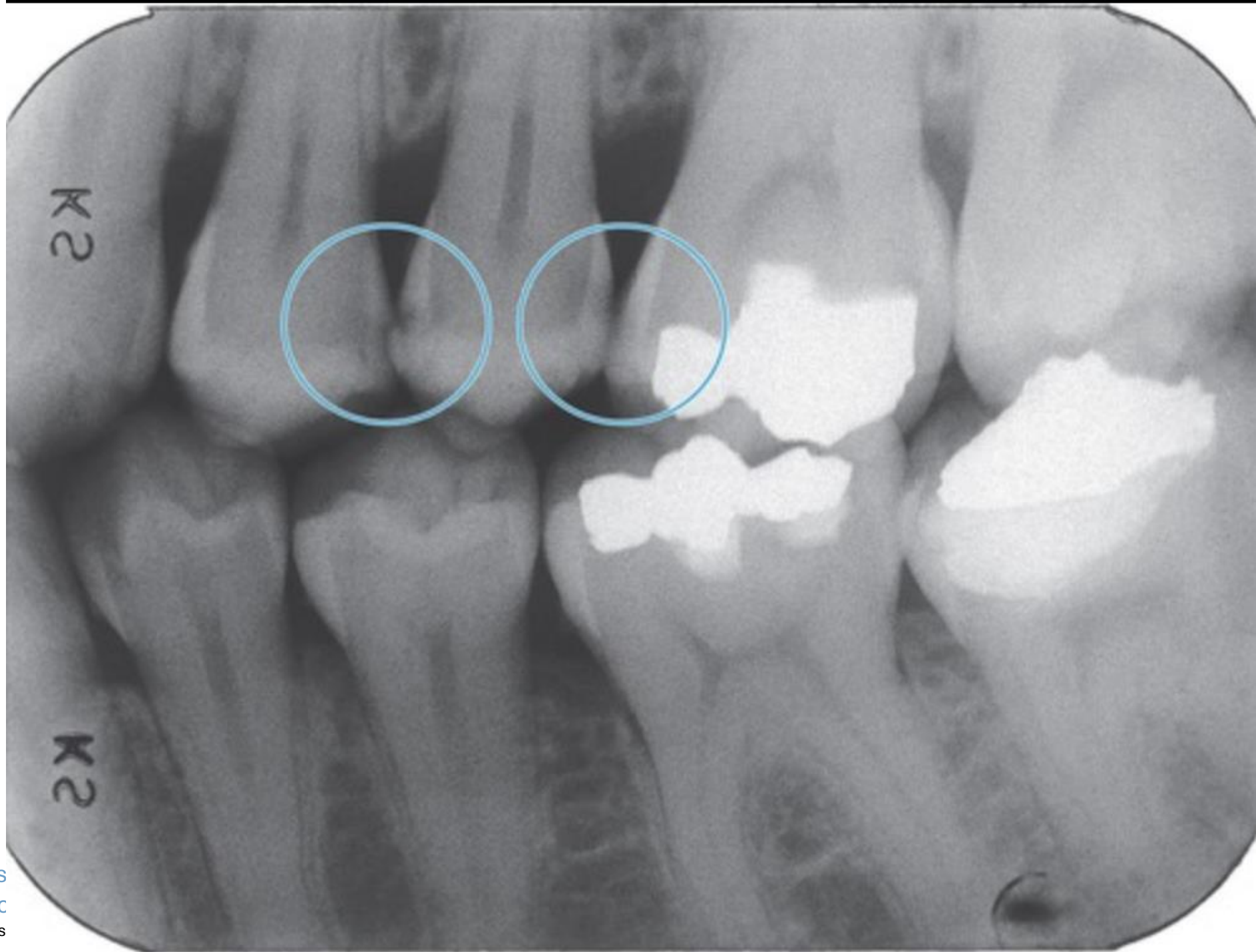
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Left Premolar Bitewing

For the circled lesions:

- Identify by tooth number and surface
- Outline the lesions



LR molar PA.

#29, 30 and 31 present. 32 absent.

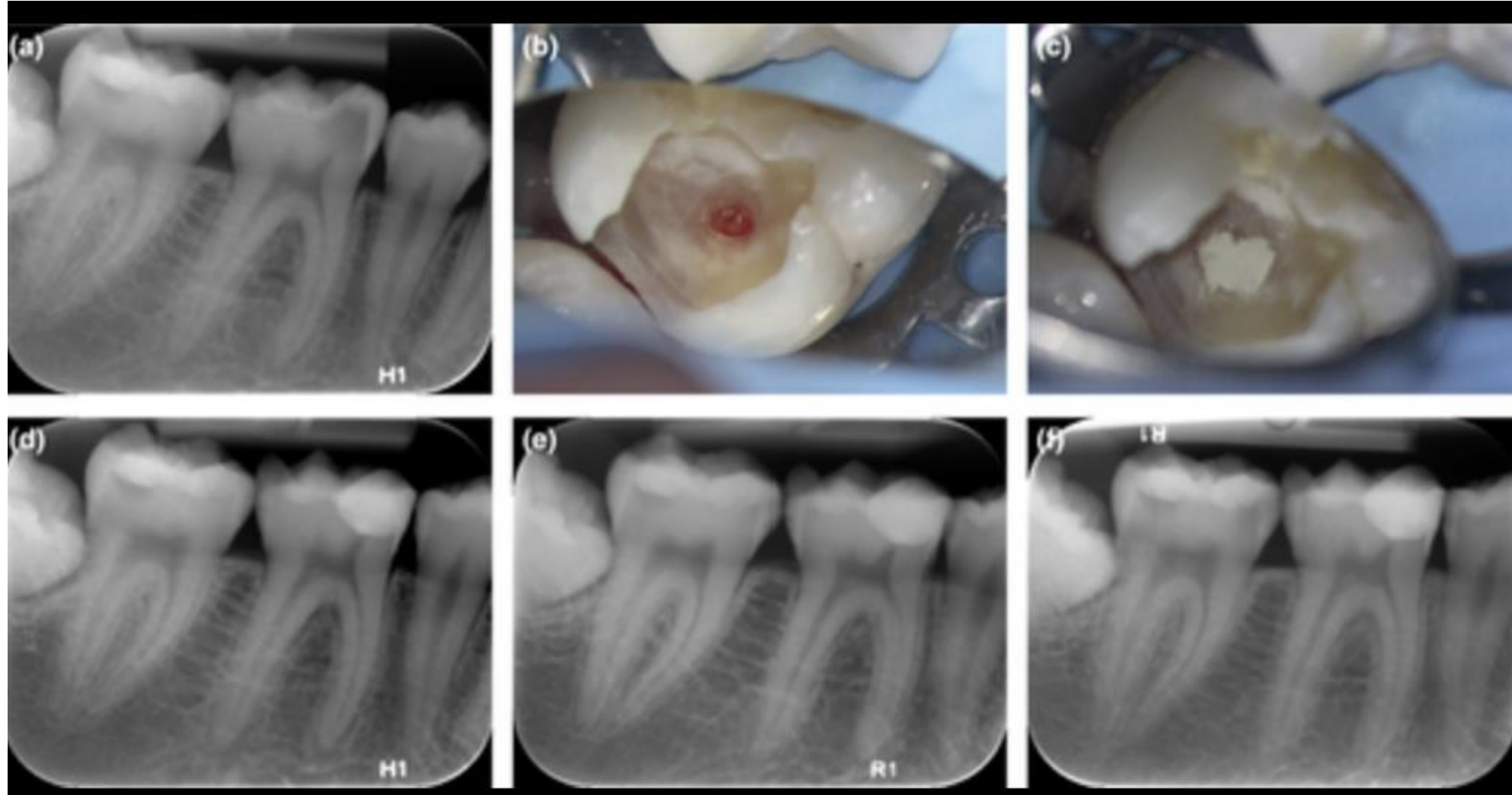
Notice:

- 30-M E2 caries
 - o 29-D? Other interproximal sites?
- 31-O D3 caries, undermining a distal cups and approximating a distal pulp horn
- Tertiary dentin formation at the distal pulp horn.



A-C: direct pulp cap
after excavation of
pulpally deep
radiographic caries on
#30-MO

D-F: Restoration is
radiographically sound.
Over time, the pdl
integrity of #30 is stable,
suggesting (but not
proving) success of the
pulp cap.
Note growth in #31
caries.

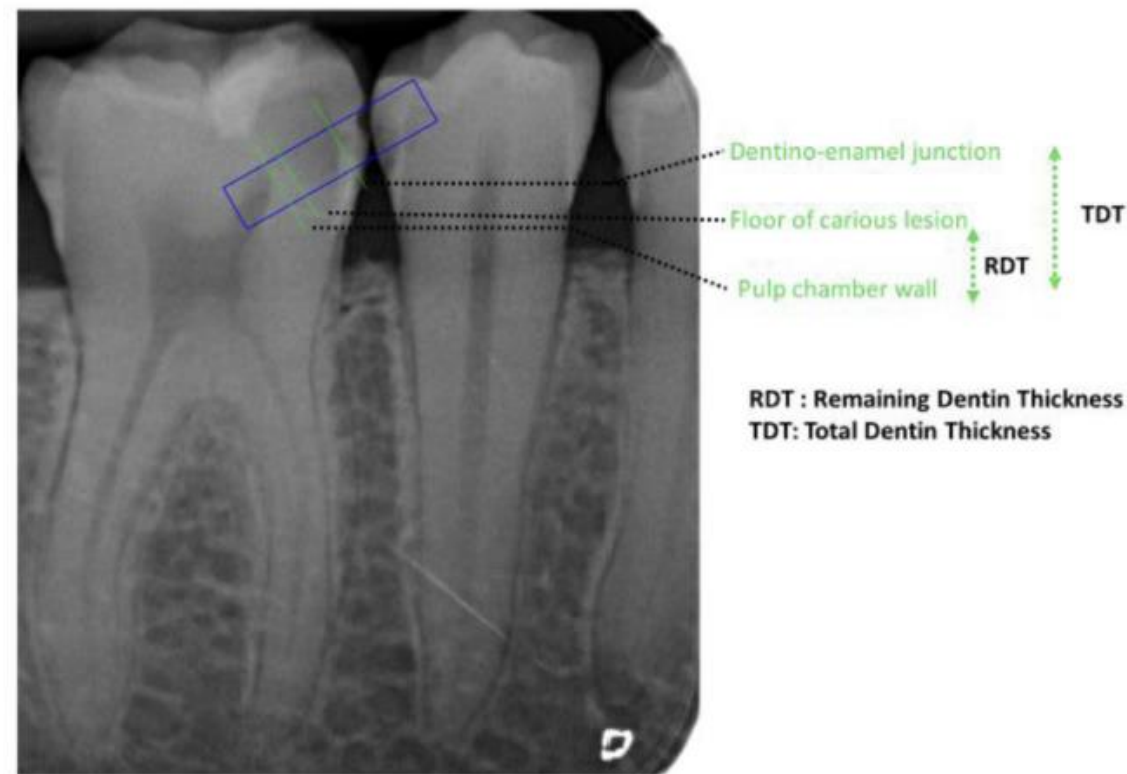


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B



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B

Can you see the extent of the dentinal lesion?

Does it undermine the cusps?

Does it have a kissing lesions?

Is this (interproximal caries) a pattern for the patient based on their habits/shared etiology?

Or is this a rare occurrence due to another factor?



Dentino-enamel junction

Floor of carious lesion

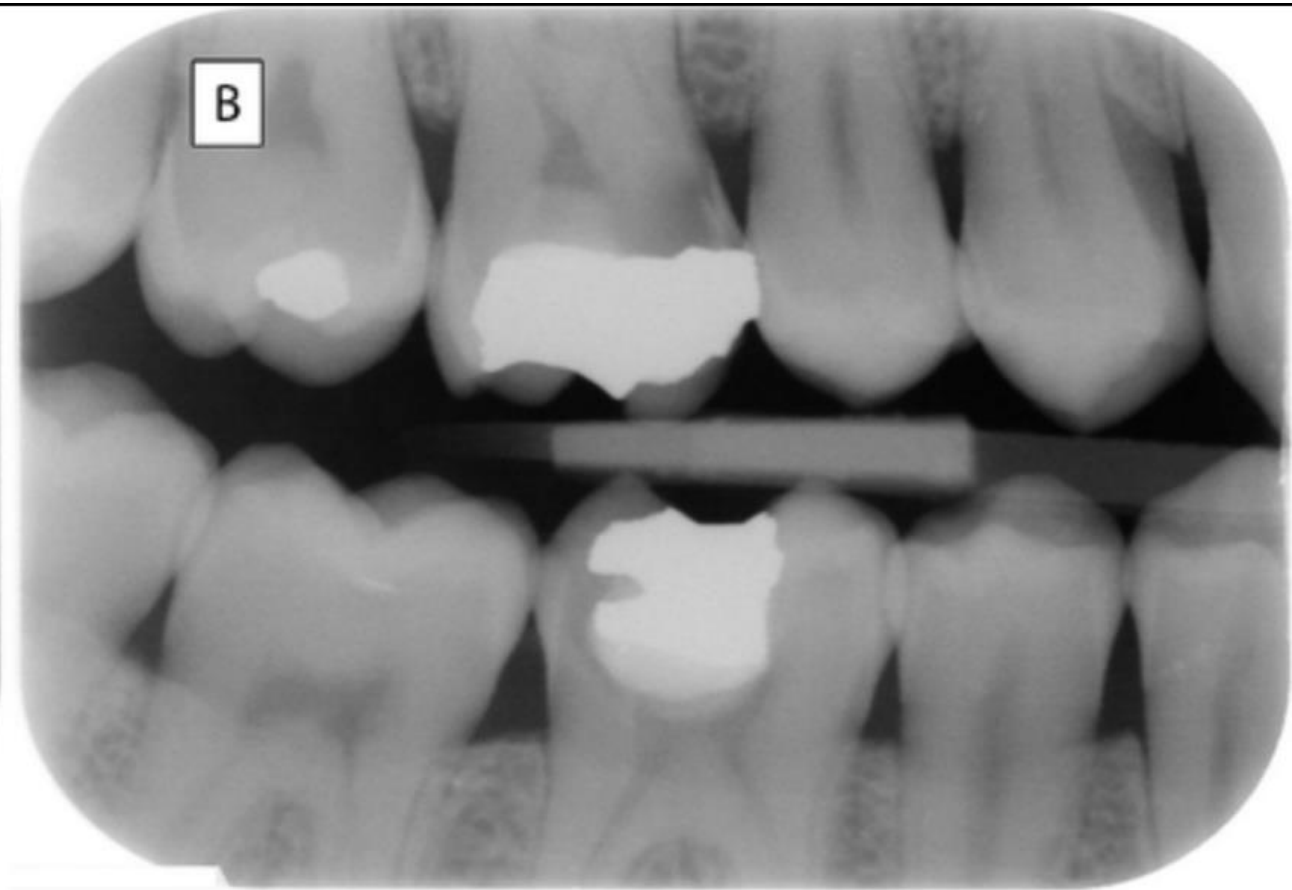
Pulp chamber wall

RDT

TDT

RDT : Remaining Dentin Thickness

TDT: Total Dentin Thickness

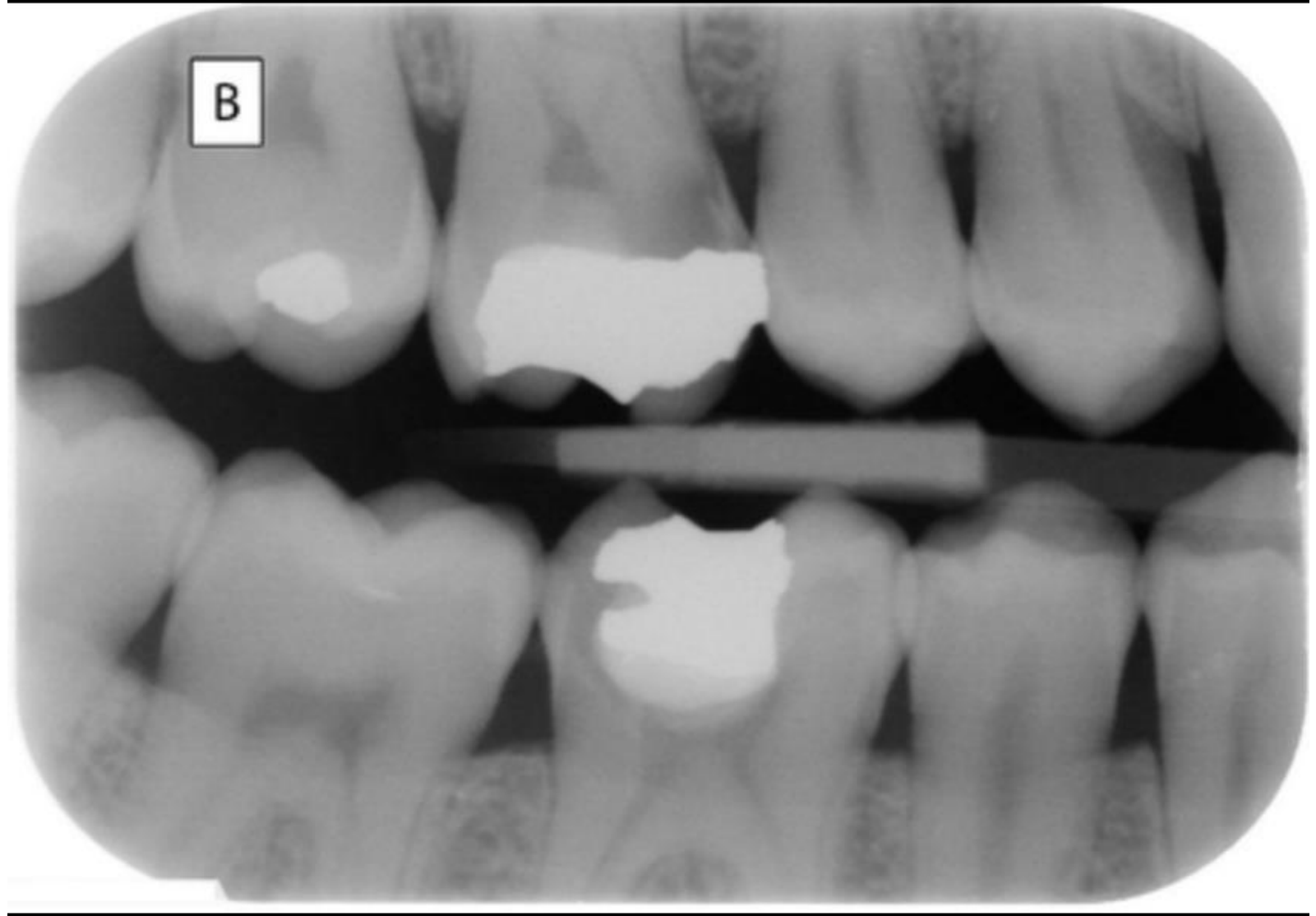


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#3:

- Describe existing
- Describe findings
- Discuss what treatments you might consider and why



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Lost Restoration



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Homework

Complete/Incomplete will affect EPA.

May be included in your future project too.

Complete your usual oral hygiene routine.

Afterwards, use a disclosing tablet to visualize residual plaque.

Take photos of your teeth

Frontal, right buccal, left buccal, occlusal of mandible (lingual of 22-27).

Photos can only be seen by grading faculty and yourself.



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This assignment does not count toward the final grade.

Lecture 3 Homework

 Publish

 Assign To

 Edit



Submissions accepted as either a .doc, .docx, or .pdf.

Failure to follow instructions or submit on time will result in a 0 for 3-24-26 EPA.

Due Date: March 31st, 2026 9:00am

1. Complete your normal oral hygiene routine at home, as described in homework assignment #1.
2. AFTER completing your routine, follow the instruction to use a disclosing tablet.
3. Take photos.
 1. REQUIRED: Frontal view
 2. Recommended: R buccal, L buccal, Occlusal view showing lingual surfaces of mandibular anterior teeth.

Points 0

Submitting a file upload

File Types doc, docx, and pdf

Due	For	Available from	Until
Mar 31 at 9am	Everyone	-	Mar 31 at 9am

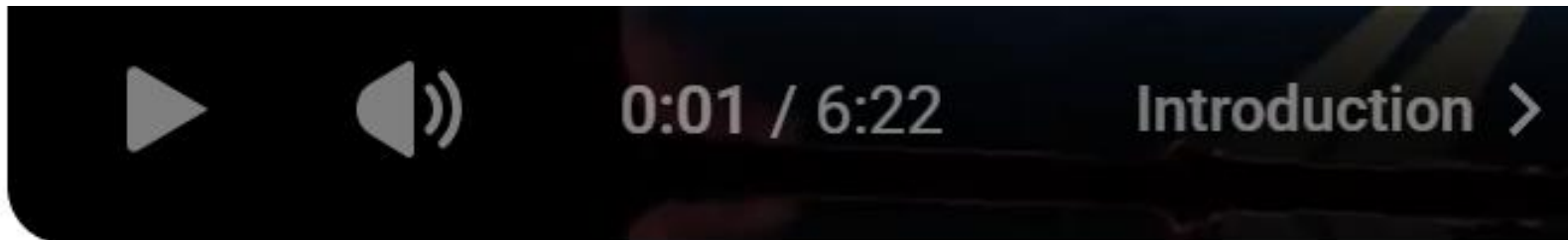
[accountability](#)

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Homework

Watch this video on youtube.

https://www.youtube.com/watch?v=IL_CuF45yY0&t=19s



Dental Plaque and Caries



Hack Dentistry

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